

JESS Thermodynamic Database v8.9

Antimony

24-Jan-23

Reaction No. 2563							
$\text{Sb+3(2)}_{\text{S-2(3)}}(\text{s}) + \text{H+1(2)}_{\text{S-2}} = \text{Sb+3(2)}_{\text{S-2(4)}} + 2\text{<H+1>}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: -19.58(4\text{SF})$	1	ELC	
2	25	0.1	Mixed salts	$\lg K: -19.58(0.15\text{SD})$	0	MSL	14007
Note (2): I not constant, I values given in paper							
3	90	0.1	Mixed salts	$\lg K: -17.44(0.13\text{SD})$	0	MSL	14007
Note (3): I not constant, I values given in paper							
4	200	0.1	Mixed salts	$\lg K: -16.98(0.10\text{SD})$	0	MSL	14007
Note (4): I not constant, I values given in paper							

Reaction No. 2732							
$\text{Sb+3(2)}_{\text{S-2(3)}}(\text{s}) + \text{H+1(2)}_{\text{S-2}} = \text{Sb+3(2)}_{\text{H+1 S-2(4)}} + \text{H+1}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: -8.5(2\text{SF})$	1	ELC	
2	25	0.1	Mixed salts	$\lg K: -10.06(0.24\text{SD})$	0	MSL	14007
Note (2): I not constant, I values given in paper							
3	90	0.1	Mixed salts	$\lg K: -9.02(0.18\text{SD})$	0	MSL	14007
Note (3): I not constant, I values given in paper							
4	200	0.1	Mixed salts	$\lg K: -9.41(0.09\text{SD})$	0	MSL	14007
Note (4): I not constant, I values given in paper							
5	275	0.1	Mixed salts	$\lg K: -9.48(0.41\text{SD})$	0	MSL	14007
Note (5): I not constant, I values given in paper							
6	350	0.1	Mixed salts	$\lg K: -9.55(0.15\text{SD})$	0	MSL	14007
Note (6): I not constant, I values given in paper							

Reaction No. 2733							
$\text{Sb+3(2)}_{\text{S-2(3)}}(\text{s}) + \text{H+1(2)}_{\text{S-2}} = \text{Sb+3(2)}_{\text{H+1(2)}_{\text{S-2(4)}}}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: -5.155(4\text{SF})$	3	EOD	31708
2	25	0.1	Mixed salts	$\lg K: -5.15(0.10\text{SD})$	2	MSL	14007
Note (2): I not constant, I values given in paper							

3	50	0	Inf. Dilution	lgK:-4.644(4SF)	2	EOD	31708
4	90	0.1	Mixed salts	lgK:-5.46(0.26SD)	0	MSL	14007
Note (4): I not constant, I values given in paper							
5	100	0	Inf. Dilution	lgK:-4.067(4SF)	0	EOD	31708
6	150	0	Inf. Dilution	lgK:-3.851(4SF)	0	EOD	31708
7	200	0	Inf. Dilution	lgK:-3.837(4SF)	0	EOD	31708
8	200	0.1	Mixed salts	lgK:-3.77(0.16SD)	0	MSL	14007
Note (8): I not constant, I values given in paper							
9	250	0	Inf. Dilution	lgK:-3.870(4SF)	0	EOD	31708
10	300	0	Inf. Dilution	lgK:-4.007(4SF)	0	EOD	31708
11	350	0	Inf. Dilution	lgK:-3.818(4SF)	0	EOD	31708

Reaction No. 2734							
$2\text{Sb}+5\text{OH}^{-1}(6) + 2\text{H}^{+1} = \text{Sb}_2\text{O}_5(\text{s}) + 7\text{H}_2\text{O}$							
Note: Sb ₂ O ₅ (s) very probably not present in original experiment							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:17.7(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:-4.16(3SF)	0	EFE	13986
3	25	0	H ₂ SO ₄	lgK:11.7(3SF)	0	MSL	23088
Note (3): Conflict between measurements in acid & alkali; this deprecated							

Reaction No. 2735							
$\text{Sb}_2\text{O}_5(\text{s}) + 2\text{H}^{+1} + 3\text{H}_2\text{O} = 2\text{Sb}+5\text{OH}^{-1}(4)$							
Note: Sb ₂ O ₅ (s) very probably not present in original experiment; does Sb(OH) ₄ exist?							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-8.4(2SF)	0	ELC	
2	25	0	Inf. Dilution	lgK:-4.70(3SF)	0	EFE	13986

Reaction No. 3604							
$\text{Sb}+3\text{Tiron-4}(2) + \text{H}^{+1} + \text{H}_2\text{O} = \text{Sb}+3\text{OH}^{-1}\text{Tiron-4} + \text{H}^{+1}(2)\text{Tiron-4}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	NaClO ₄	lgK:1.23(3SF)	6	ENS	772
2	25	0	Inf. Dilution	lgK:1.23(3SF)	1	ESC	

Reaction No. 4873							
$0.5\text{Sb}+3(2)\text{S-2}(3)\text{(s)} + 3\text{H}^{+1} + 4\text{Cl}^{-1} = \text{Sb}+3\text{Cl}^{-1}(4) + 1.5\text{H}^{+1}(2)\text{S-2}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

1	25	0	Inf. Dilution	lgK:-11.7 (3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:-12.24 (4SF)	0	MSL	140
3	25	0	Inf. Dilution	lgK:-12.24 (4SF)	0	MSL	14015
Note (3): Inconsistent with MPL data for Sb+3_Cl-(4)							

Reaction No. 5143							
Sb+3(2)_H+1(-2)_OH-1(2)_Tartaric-2(2) + 2<OH-1> = 2<H2O> + 2<Sb+3_H+1(-2)_OH-1_Tartaric-2 >							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	NaClO4	lgK:10.16 (4SF)	5	MGL	772
Note (1): both D ad L same values; meso completely different response							
2	25	0	NaClO4	lgK:10.16 (3SF)	1	ESC	

Reaction No. 7217							
Sb+3_dldMSA-4(2) + H2O = Sb+3_OH-1_dldMSA-4(2) + H+1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	NaClO4	lgK:-4.90 (3SF)	6	MGL	772
2	25	0	Inf. Dilution	lgK:-4.90 (3SF)	1	ESC	

Reaction No. 7351							
Sb+3_OH-1_2SHAcet-2(2) + H+1 = Sb+3_2SHAcet-2(2) + H2O							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	NaClO4	lgK:7.58 (3SF)	6	MGL	772
2	25	0	Inf. Dilution	lgK:7.58 (3SF)	1	ESC	

Reaction No. 7352							
Sb+3_2SHAcet-2(2) + 3<OH-1> = Sb+3_OH-1(3) + 2<2SHAcet-2>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	NaClO4	lgK:6.92 (3SF)	6	MGL	772
2	25	0	Inf. Dilution	lgK:6.9 (2SF)	1	ESC	

Reaction No. 10638							
Sb+3(2)_H+1(-2)_OH-1(2)_Tartaric-2(2) + 2<OH-1> + 2<H2O> = 2<Sb+3_OH-1(3)> + 2<Tartaric-2>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	NaClO4	lgK:6.07 (3SF)	5	MGL	772
Note (1): both D ad L same values; meso completely different response							
2	25	0	Inf. Dilution	lgK:6.1 (2SF)	1	ESC	

Reaction No. 10639							
$\text{Sb+3_OH-1(3)} + 2\text{<H+1_2SHEtol-1>} = \text{Sb+3_H+1(-2)_2SHEtol-1(2)} + 3\text{<H2O>} + \text{H+1}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	NaClO4	lgK:-1.73(3SF)	6	MGL	772
2	25	0	Inf. Dilution	lgK:-1.73(3SF)	1	ESC	

Reaction No. 10987							
$\text{Sb+3_H+1_2SHSuccin-3(2)} + \text{H+1} = \text{Sb+3_H+1(2)_2SHSuccin-3(2)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	NaClO4	lgK:2.4(2SF)	4	MGL	772
2	25	0	Inf. Dilution	lgK:2.4(2SF)	1	ESC	

Reaction No. 10988							
$\text{Sb+3_2SHSuccin-3(2)} + \text{H+1} = \text{Sb+3_H+1_2SHSuccin-3(2)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	NaClO4	lgK:3.46(3SF)	6	MGL	772
2	25	0	Inf. Dilution	lgK:3.46(3SF)	1	ESC	

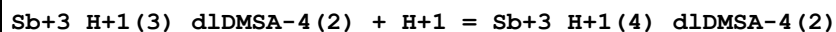
Reaction No. 10989							
$\text{Sb+3_2SHSuccin-3(2)} + 3\text{<OH-1>} = \text{Sb+3_OH-1(3)} + 2\text{<2SHSuccin-3>}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	NaClO4	lgK:5.90(3SF)	6	MGL	772
2	25	0	Inf. Dilution	lgK:5.9(2SF)	1	ESC	

Reaction No. 10999							
$2\text{<Sb+3_OH-1_DMSA-4>} + 2\text{<H+1>} = \text{Sb+3(2)_DMSA-4(2)} + 2\text{<H2O>}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	NaClO4	lgK:13.17(4SF)	6	MGL	772
Note (1): meso form; there are values also for dl							
2	25	0	Inf. Dilution	lgK:13.17(4SF)	1	ESC	

Reaction No. 11006							
$\text{Sb+3(2)_DMSA-4(2)} + 6\text{<OH-1>} = 2\text{<Sb+3_OH-1(3)>} + 2\text{<DMSA-4>}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	NaClO4	lgK:9.2(2SF)	4	MGL	772
Note (1): meso form; there are values also for dl							

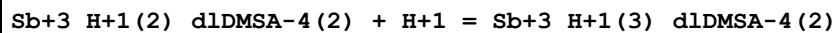
2	25	0	Inf. Dilution	lgK:9.2 (2SF)	1	ESC	
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Reaction No. 11019



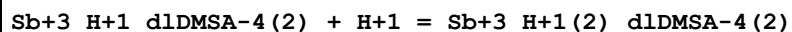
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	NaClO4	lgK:2.57 (3SF)	6	MGL	772
2	25	0	Inf. Dilution	lgK:2.57 (3SF)	1	ESC	

Reaction No. 11020



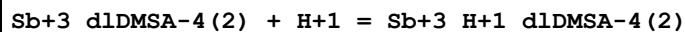
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	NaClO4	lgK:3.60 (3SF)	6	MGL	772
2	25	0	Inf. Dilution	lgK:3.60 (3SF)	1	ESC	

Reaction No. 11021



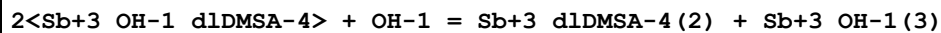
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	NaClO4	lgK:4.61 (3SF)	6	MGL	772
2	25	0	Inf. Dilution	lgK:4.61 (3SF)	1	ESC	

Reaction No. 11022



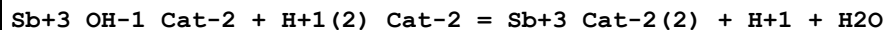
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	NaClO4	lgK:6.82 (3SF)	6	MGL	772
2	25	0	Inf. Dilution	lgK:6.82 (3SF)	1	ESC	

Reaction No. 11023



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	NaClO4	lgK:-10.7 (3SF)	6	MGL	772
2	25	0	Inf. Dilution	lgK:-10.7 (3SF)	1	ESC	

Reaction No. 11806



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	NaClO4	lgK:-2.37 (3SF)	6	MGL	772
2	25	0	Inf. Dilution	lgK:-2.37 (3SF)	1	ESC	

Reaction No. 11807							
$\text{Sb+3_OH-1(3)} + 2\text{<H+1_Cat-2>} = \text{Sb+3_Cat-2(2)} + \text{OH-1} + 2\text{<H2O>}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	NaClO4	lgK:5.44(3SF)	6	MGL	772
2	25	0	Inf. Dilution	lgK:5.44(3SF)	1	ESC	

Reaction No. 12487							
$\text{Sb+3_Tiron-4} + \text{H+1(2)_Tiron-4} = \text{Sb+3_Tiron-4(2)} + 2\text{<H+1>}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-5.73(3SF)	1	ESC	
2	25	0.1	KNO3	lgK:-5.73(0.01SD)	3	MGL	5664
3	25	1	KCl	lgK:-6.01(0.06SD)	3	MGL	5664

Reaction No. 12488							
$\text{Sb+3_Tiron-4(2)} + \text{OH-1} + 2\text{<H2O>} = \text{Sb+3_OH-1(3)} + 2\text{<H+1_Tiron-4>}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	NaClO4	lgK:-3.95(3SF)	6	MGL	772
2	25	0	Inf. Dilution	lgK:-3.95(3SF)	1	ESC	

Reaction No. 12489							
$\text{Sb+3_H+1_Tiron-4(2)} = \text{Sb+3_Tiron-4(2)} + \text{H+1}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-2.00(3SF)	1	ESC	
2	25	0.1	KNO3	lgK:-2.00(3SF)	4	MGL	5664
3	25	1	KCl	lgK:-2.04(3SF)	4	MGL	5664

Reaction No. 12491							
$\text{Sb+3_Tiron-4} + \text{Tiron-4} = \text{Sb+3_Tiron-4(2)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	KNO3	lgK:14.5(3SF)	1	MDG	5664
Note (1): Just combination of other measured constants							

Reaction No. 13005							
$\text{Sb+3_23DHB-3(2)} + \text{OH-1} + 2\text{<H2O>} = \text{Sb+3_OH-1(3)} + 2\text{<H+1_23DHB-3>}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	NaClO4	lgK:-4.17(3SF)	6	MGL	772
2	25	0	Inf. Dilution	lgK:-4.2(2SF)	1	ESC	

Reaction No. 13006							
$\text{Sb+3_23DHB-3(2)} + \text{H+1} + \text{H2O} = \text{Sb+3_OH-1_23DHB-3} + \text{H+1(2)_23DHB-3}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	NaClO ₄	lgK:2.8(2SF)	4	MGL	772
2	25	0	Inf. Dilution	lgK:3.0(2SF)	1	ESC	

Reaction No. 14017							
$\text{Sb+3_EDTA-4} + 2\text{<OH-1>} + \text{H2O} = \text{Sb+3_OH-1(3)} + \text{H+1_EDTA-4}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	NaClO ₄	lgK:12.46(4SF)	6	EIU	903
2	20	0.1	NaClO ₄	lgK:12.46(0.05SD)	4	MGL	2038
3	20	0.1	Unknown	lgK:12.46(4SF)	5	CRV	7271
4	25	0	Inf. Dilution	lgK:11.8(3SF)	2	EAE	

Reaction No. 14018							
$\text{Sb+3_EDTA-4} + \text{H+1} = \text{Sb+3_H+1_EDTA-4}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	Unknown	lgK:1.0(2SF)	4	CRV	7271
2	25	0.1	KNO ₃	lgK:1.02(3SF)	2	MGL	5664

Reaction No. 14019							
$\text{Sb+3_EDTA-4} + \text{OH-1} = \text{Sb+3_OH-1_EDTA-4}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	KNO ₃	lgK:8.24(0.05SD)	5	MGL	5664
2	25	0.1	Unknown	lgK:8.24(3SF)	4	CRV	7271
3	25	1	NaClO ₄	lgK:-8.7(2SF)	0	MSP	931
4	25	1	NaClO ₄	lgK:8.69(3SF)	5	MSP	14056
5	25	1	Unknown	lgK:8.69(3SF)	4	CRV	7271

Reaction No. 14020							
$\text{Sb+3_OH-1(2)_EDTA-4} + \text{H+1} = \text{Sb+3_OH-1_EDTA-4} + \text{H2O}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:7.46(3SF)	1	ESC	
2	25	0.1	KNO ₃	lgK:7.46(3SF)	3	MGL	5664
Note (2): Sign changed							

Reaction No. 14132							
Sb+3_HOEDTA-3 + H2O = Sb+3_OH-1_HOEDTA-3 + H+1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	KNO3	lgK:-3.2 (2SF)	4	MEF	999
2	20	0.1	NaClO4	lgK:-3.1 (2SF)	6	MGL	2038
3	20	0.1	Unknown	lgK:-3.1 (2SF)	5	CRV	552
Note (3): Origin Anderegg HCA, 70, 53, 564 [2038]							
4	25	0.1	KNO3	lgK:-3.05 (0.05SD)	6	MGL	5664
5	25	1	NaClO4	lgK:-2.80 (3SF)	3	MGL	14055
6	25	1	NaClO4	lgK:-3.15 (3SF)	5	MSP	14055
7	25	1	Unknown	lgK:-3.2 (2SF)	5	CRV	552
Note (7): Origin Bhat et al. Z. Anorg. Chem., 1966, 343, 329 [14055]							

Reaction No. 14747							
Sb+3_H+1_DTPA-5 = Sb+3_DTPA-5 + H+1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	NaClO4	lgK:-3.57 (0.05SD)	4	MGL	772
2	20	0.1	NaClO4	lgK:-3.57 (0.05SD)	5	MGL	2038
3	20	0.1	NaClO4	lgK:-3.57 (0.05SD)	5	CRV	13242
4	25	0.1	KNO3	lgK:-3.31 (0.04SD)	5	MGL	5664

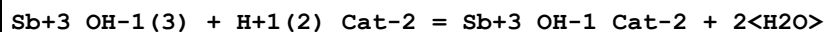
Reaction No. 19804							
Sb+3_DTPA-5 + 2<OH-1> + H2O = Sb+3_OH-1(3) + H+1_DTPA-5							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	NaClO4	lgK:9.82 (0.05SD)	6	MGL	2038
2	25	0	Inf. Dilution	lgK:9.8 (2SF)	1	ESC	

Reaction No. 20744							
Sb+3_OH-1(3) + H+1(3)_CDTA-4 = Sb+3_CDTA-4 + 3<H2O>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	Unknown	lgK:29.5 (3SF)	0	ENS	773
Note (1): Not as published, unknown procedure to reach this logK value							

Reaction No. 20747							
Sb+3_OH-1(3) + H+1_Cat-2 + H+1(2)_Cat-2 = Sb+3_Cat-2(2) + 3<H2O>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	Unknown	lgK:10.2 (3SF)	0	ENS	773

Note (1): Not as published, unknown procedure to reach this logK value

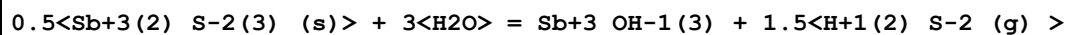
Reaction No. 20748



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	Unknown	lgK:3.3 (2SF)	0	ENS	773

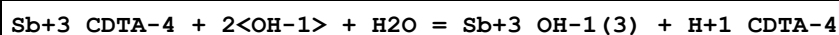
Note (1): Not as published, unknown procedure to reach this logK value

Reaction No. 21384



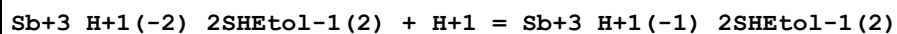
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-18.8 (3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:-14.7 (3SF)	0	EFE	140

Reaction No. 24372



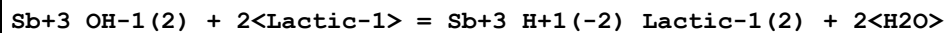
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	NaClO4	lgK:11.2 (0.05SD)	4	MGL	772
2	25	0	Inf. Dilution	lgK:11.2 (3SF)	1	ESC	

Reaction No. 24466



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	NaClO4	lgK:7.98 (3SF)	6	MGL	772
2	25	0	Inf. Dilution	lgK:7.98 (3SF)	1	ESC	

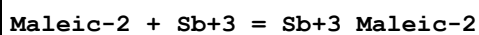
Reaction No. 24496



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:7.7 (2SF)	1	ESC	
2	35	0	Unknown	lgK:7.7 (2SF)	0	MSL	22955

Note (2): I not kept constant, no electrolyte added; Sb(OH)2 not main species at working pH

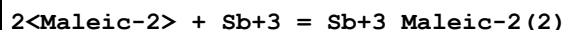
Reaction No. 31006



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.7	NaClO4	lgK:7.78 (3SF)	0	MPL	3009

Note (1): Sb3+ is not what the authors had in solution

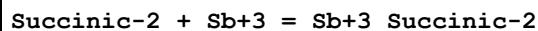
Reaction No. 31007



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.7	NaClO4	lgK:14.95(4SF)	0	MPL	3009

Note (1): Sb3+ is not what the authors had in solution

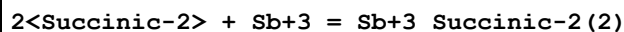
Reaction No. 31045



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.7	NaClO4	lgK:8.70(3SF)	0	MPL	3009

Note (1): Sb3+ is not what the authors had in solution

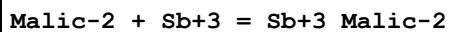
Reaction No. 31046



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.7	NaClO4	lgK:17.60(4SF)	0	MPL	3009

Note (1): Sb3+ is not what the authors had in solution

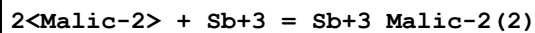
Reaction No. 31054



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.7	NaClO4	lgK:8.54(3SF)	0	MPL	3009

Note (1): Sb3+ is not what the authors had in solution

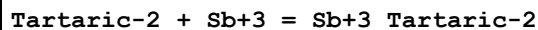
Reaction No. 31055



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.7	NaClO4	lgK:17.18(4SF)	0	MPL	3009

Note (1): Sb3+ is not what the authors had in solution

Reaction No. 31062



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.7	NaClO4	lgK:8.08(3SF)	0	MPL	3009

Note (1): Sb3+ is not what the authors had in solution

Reaction No. 31063							
2<Tartaric-2> + Sb+3 = Sb+3_Tartaric-2(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.7	NaClO4	lgK:14.81(4SF)	0	MPL	3009
Note (1): Sb3+ is not what the authors had in solution							

Reaction No. 31940							
Glutaric-2 + Sb+3 = Sb+3_Glutaric-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.7	NaClO4	lgK:9.48(3SF)	0	MPL	3009
Note (1): Sb3+ is not what the authors had in solution							

Reaction No. 31941							
2<Glutaric-2> + Sb+3 = Sb+3_Glutaric-2(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.7	NaClO4	lgK:18.78(4SF)	0	MPL	3009
Note (1): Sb3+ is not what the authors had in solution							

Reaction No. 33505							
Formic-1 + Sb+3 = Sb+3_Formic-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.7	NaClO4	lgK:4.60(3SF)	0	MGL	3009
Note (1): Sb3+ is not what the authors had in solution							

Reaction No. 33506							
2<Formic-1> + Sb+3 = Sb+3_Formic-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.7	NaClO4	lgK:9.53(3SF)	0	MGL	3009
Note (1): Sb3+ is not what the authors had in solution							

Reaction No. 33507							
Sb+3 + Propanoic-1 = Sb+3_Propanoic-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.7	NaClO4	lgK:6.68(3SF)	0	MGL	3009
Note (1): Sb3+ is not what the authors had in solution							

Reaction No. 33508							
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2<Propanoic-1> + Sb+3 = Sb+3_Propanoic-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.7	NaClO4	lgK:11.20(4SF)	0	MGL	3009
Note (1): Sb3+ is not what the authors had in solution							

Reaction No. 33509							
Lactic-1 + Sb+3 = Sb+3_Lactic-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.7	NaClO4	lgK:7.84(3SF)	0	MGL	3009
Note (1): Sb3+ is not what the authors had in solution							

Reaction No. 33510							
2<Lactic-1> + Sb+3 = Sb+3_Lactic-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.7	NaClO4	lgK:12.00(4SF)	0	MGL	3009
Note (1): Sb3+ is not what the authors had in solution							

Reaction No. 33511							
Malonic-2 + Sb+3 = Sb+3_Malonic-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.7	NaClO4	lgK:10.18(4SF)	0	MGL	3009
Note (1): Sb3+ is not what the authors had in solution							

Reaction No. 33512							
2<Malonic-2> + Sb+3 = Sb+3_Malonic-2(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.7	NaClO4	lgK:16.34(4SF)	0	MGL	3009
Note (1): Sb3+ is not what the authors had in solution							

Reaction No. 33811							
Acetic-1 + Sb+3 = Sb+3_Acetic-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.7	NaClO4	lgK:7.00(3SF)	0	MPL	3009
Note (1): Sb3+ is not what the authors had in solution							

Reaction No. 33812							
2<Acetic-1> + Sb+3 = Sb+3_Acetic-1(2)							

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.7	NaClO4	lgK:12.64(4SF)	0	MPL	3009
Note (1): Sb3+ is not what the authors had in solution							

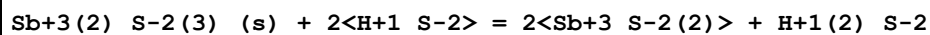
Reaction No. 34202							
Sb+3_H+1(-1)_DDC-1_Dithizone-1 + 2<H+1_DDC-1> = Sb+3_DDC-1(3) + H+1_Dithizone-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20		CCl4	lgK:2.5(0.07SD)	5	MSP	931

Reaction No. 35374							
0.5<O2(g)> + Sb(s) - e-1 = SbO+1							
Note: SbO+1 is not considered to exist in aqueous solution. It would be hydrated. The predominant equivalent species is Sb+3_OH-1(4). However, SbO+1 is left here because it appears elsewhere. This should be OK as long as this (single) reference equilibrium operates everywhere the same. All w 0, otherwise interferes with calculations							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-42.0(3SF)	0	EES	
2	25	0	Inf. Dilution	dG:-177.(3SF)kJ	0	CRV	7421
3	25	0	Inf. Dilution	dG:-177.11(5SF)kJ	0	EFE	9015
4	25	0	Inf. Dilution	dG:-42.33(4SF)	4	CRV	12011
5	25	0	Inf. Dilution	dG:-42.0(3SF)	0	EFE	13986
Note (5): Copied from NBS1952 [15886]							
6	25	0	Inf. Dilution	dG:-42.0(3SF)	0	EFE	15886
7	25	0	Inf. Dilution	dG:-42.33(4SF)	0	EFE	23127
Note (7): Copied from Wagman1982 [9015]							
8	25	0	Inf. Dilution	dG:-177.11(5SF)kJ	0	EOD	29489
Note (8): Copied from Wagman1982 [9015]							
9	25	0	Inf. Dilution	dG:-42.0(3SF)	0	EFE	31675
10	25	0	Inf. Dilution	dG:-42.33(4SF)	0	EFE	31675

Reaction No. 36006							
Sb+3(2)_S-2(3)_(s) + H+1_S-2 = Sb+3(2)_H+1_S-2(4)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-1.54(3SF)	1	EES	
2	25	0	Inf. Dilution	lgK:-2.33(3SF)	0	MSL	140
Note (2): Based on Akeret 53							
3	25	0	Inf. Dilution	lgK:-2.50(0.2SD)	0	EOD	14006

4	25	0	Inf. Dilution	lgK:-2.893(4SF)	0	EOD	31708
5	30	0	Self Medium	lgK:-5.4(0.2SD)	0	MSL	29949
6	50	0	Inf. Dilution	lgK:-1.80(0.2SD)	0	EOD	14006
7	50	0	Inf. Dilution	lgK:-2.656(4SF)	0	EOD	31708
8	100	0	Inf. Dilution	lgK:-0.80(0.2SD)	0	EOD	14006
9	100	0	Inf. Dilution	lgK:-2.521(4SF)	0	EOD	31708
10	150	0	Inf. Dilution	lgK:-0.40(0.2SD)	0	EOD	14006
11	150	0	Inf. Dilution	lgK:-2.643(4SF)	0	EOD	31708
12	200	0	Inf. Dilution	lgK:-0.25(0.2SD)	0	EOD	14006
13	200	0	Inf. Dilution	lgK:-2.897(4SF)	0	EOD	31708
14	250	0	Inf. Dilution	lgK:-0.15(0.2SD)	0	EOD	14006
15	250	0	Inf. Dilution	lgK:-3.258(4SF)	0	EOD	31708
16	300	0	Inf. Dilution	lgK:-0.10(0.2SD)	0	EOD	14006
17	300	0	Inf. Dilution	lgK:-3.538(4SF)	0	EOD	31708
18	350	0	Inf. Dilution	lgK:-3.759(4SF)	0	EOD	31708

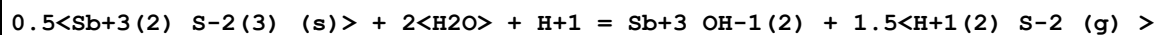
Reaction No. 36007



Note: Species discredited in later evaluation of data from this publication

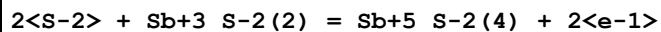
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-2.03(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:-2.03(3SF)	0	MSL	14015

Reaction No. 36069



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-17.5(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:-13.9(3SF)	0	EFE	140

Reaction No. 36070



Note: The species S-2 does not exist in aqueous solution See reaction: H+1 + S-2 = H+1S-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:0.6(1SF)	0	EFE	600

Reaction No. 36071

$\text{Sb(s)} + 2\text{S-2} = \text{Sb+3}_{\text{S-2}}(2) + 3\text{e-1}$ Note: The species S-2 does not exist in aqueous solution See reaction: $\text{H+1} + \text{S-2} = \text{H+1}_{\text{S-2}}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:38.0 (3SF)	0	ELC	
2	25	0	Inf. Dilution	E:0.85 (2SF)	0	EFE	600

Reaction No. 36072							
$2\text{Sb+3}_{\text{OH-1}}(2) + 3\text{H+1}_{\text{S-2}}(\text{g}) = \text{Sb+3}(2)_{\text{S-2}}(3) + 4\text{H}_2\text{O} + 2\text{H+1}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:35.0 (3SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-38. (2SF)	0	EFE	600
Note (2): Estimated from an estimation of entropy not justified and "a heat of formation "of orange pr.							

Reaction No. 36073							
$\text{Sb+3}_{\text{Cl-1}}(3) + 2\text{H}_2\text{O} = \text{Sb+3}_{\text{OH-1}}(2) + 2\text{H+1} + 3\text{Cl-1}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.4 (2SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-1.500 (4SF)	0	EFE	600
Note (2): SbCl3(s) very soluble; sol. never determined; est. from old DG data							

Reaction No. 43844							
$\text{EDTA-4} + \text{Sb+3} = \text{Sb+3}_{\text{EDTA-4}}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	4	NaCl	lgK:19.48 (4SF)	0	MPL	3459
Note (1): Sb+3 not what the authors had in solution							
2	25	4	NaCl	dG:-51.537 (5SF) kJ	0	EFE	3459
Note (2): Inconsistent even with lgK from same study							
3	25	4	NaCl	dH:-25.140 (5SF) kJ	0	MCL	3459

Reaction No. 43845							
$\text{CDTA-4} + \text{Sb+3} = \text{Sb+3}_{\text{CDTA-4}}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	4	NaCl	lgK:24.8 (3SF)	0	MPL	3459
Note (1): Sb+3 not what the authors had in solution							
2	25	4	NaCl	dG:-34.048 (5SF)	0	EFE	3459
3	25	4	NaCl	dH:-17.024 (5SF)	0	MCL	3459

Reaction No. 45972							
Sb+3_OH-1_HOEDTA-3 + OH-1 = Sb+3_OH-1(2)_HOEDTA-3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:8.1(2SF)	1	ESC	
2	25	1	NaClO4	lgK:8.13(3SF)	4	MGL	14055
3	25	1	NaClO4	lgK:8.11(3SF)	4	MSP	14055
4	25	1	Unknown	lgK:8.13(3SF)	4	CRV	2556
Note (4): NIST Stab. Const. Database V5 shows this also							

Reaction No. 53125							
Sb+3 + Cl-1 = Sb+3_Cl-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	4.7	(H,Na)ClO4	lgK:1.05(3SF)	0	MPL	5296
Note (1): Grossly inconsistent							
2	22	4	(H,Na)ClO4	lgK:0.96(2SF)	0	MPL	13994
Note (2): Grossly inconsistent - authors show dif. lit. values without comment							
3	25	0	Inf. Dilution	lgK:2.3(2SF)	1	EES	
4	25	4	HClO4	lgK:2.26(3SF)	0	MPL	140
5	25	4	HClO4	lgK:2.26(3SF)	5	MPL	13998
6	30	2	NaClO4	lgK:2.30(3SF)	3	MPL	5398
Note (6): Authors take Bond Waugh 70 data but I is wrong							
7	30	5	HClO4	lgK:2.3(2SF)	3	MPL	13995
8	30	5	Unknown	lgK:2.30(3SF)	3	CRV	2556

Reaction No. 53126							
2<Cl-1> + Sb+3 = Sb+3_Cl-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	4.7	(H,Na)ClO4	lgK:1.90(3SF)	0	MPL	5296
Note (1): Grossly inconsistent							
2	22	4	(H,Na)ClO4	lgK:1.52(3SF)	0	MPL	13994
Note (2): Grossly inconsistent - authors show dif. lit. values without comment							
3	22	4	HClO4	lgK:3.39(3SF)	2	MPL	13994
4	25	0	Inf. Dilution	lgK:4.0(2SF)	1	EES	
5	25	4	HClO4	lgK:3.49(3SF)	0	MPL	140
6	25	4	HClO4	lgK:3.49(3SF)	3	MPL	13998
7	30	2	NaClO4	lgK:4.0(2SF)	3	MDG	5398

Note (7): Authors take Bond Waugh 70 data but I is wrong							
8	30	5	HClO4	lgK:4.0(2SF)	5	MPL	13995
Note (8): Two different data treatment methods gave 4.0 and 4.2							
9	30	5	Unknown	lgK:4.0(2SF)	5	CRV	2556

Reaction No. 53127							
3<Cl-1> + Sb+3 = Sb+3_Cl-1(3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	4.7	(H,Na)ClO4	lgK:2.20(3SF)	0	MPL	5296
Note (1): Grossly inconsistent							
2	22	4	(H,Na)ClO4	lgK:1.45(3SF)	0	MPL	13994
Note (2): Grossly inconsistent - authors show dif. lit. values without comment							
3	22	4	HClO4	lgK:4.09(3SF)	0	MPL	13994
4	25	0	Inf. Dilution	lgK:8.1(2SF)	1	EDH	
5	25	0	Inf. Dilution	lgK:5.8(2SF)	1	EES	
6	25	4	HClO4	lgK:4.18(3SF)	0	MPL	140
7	25	4	HClO4	lgK:4.18(3SF)	0	MPL	13998
8	30	2	NaClO4	lgK:5.8(2SF)	2	MDG	5398
Note (8): Authors take Bond Waugh 70 data but I is wrong							
9	30	5	HClO4	lgK:5.8(2SF)	4	MPL	13995
Note (9): Two different data treatment methods gave 5.8 and 6.0							
10	30	5	Unknown	lgK:5.8(2SF)	4	CRV	2556

Reaction No. 53128							
4<Cl-1> + Sb+3 = Sb+3_Cl-1(4)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	4.7	(H,Na)ClO4	lgK:1.95(3SF)	0	MPL	5296
Note (1): Grossly inconsistent							
2	22	0.8:1.2	HCl	lgK:4.7(2SF)	0	MPT	31713
3	22	4	(H,Na)ClO4	lgK:1.04(3SF)	0	MPL	13994
Note (3): Grossly inconsistent - authors show dif. lit. values without comment							
4	25	0	Inf. Dilution	lgK:8.6(2SF)	1	ELC	
5	25	4	HClO4	lgK:4.72(3SF)	0	MPL	140
6	25	4	HClO4	lgK:4.72(3SF)	0	MPL	13998
7	30	2	NaClO4	lgK:6.8(2SF)	3	MDG	5398
Note (7): Authors take Bond Waugh 70 data but it is wrong							

8	30	5	HClO ₄	lgK:6.8 (2SF)	4	MPL	13995
Note (8): Two different data treatment methods gave 6.8 and 7.1							
9	30	5	Unknown	lgK:6.8 (2SF)	4	CRV	2556

Reaction No. 53223							
Sb+3 + F-1 = Sb+3_F-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:3. (1SF)	1	EES	
2	30	2	(H,Na)ClO ₄	lgK:3.0 (0.07SD)	4	MPL	437
3	30	2	(H,Na)ClO ₄	lgK:3.00 (3SF)	5	MPL	13996
4	30	2	NaClO ₄	lgK:3.00 (3SF)	4	MPL	5398

Reaction No. 53224							
Sb+3 + 2<F-1> = Sb+3_F-1 (2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:5.70 (3SF)	1	EES	
2	30	2	(H,Na)ClO ₄	lgK:5.70 (3SF)	5	MPL	13996
3	30	2	NaClO ₄	lgK:5.70 (3SF)	4	MPL	5398

Reaction No. 53225							
Sb+3 + 3<F-1> = Sb+3_F-1 (3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:8.30 (3SF)	1	EES	
2	30	2	(H,Na)ClO ₄	lgK:8.30 (3SF)	5	MPL	13996
3	30	2	NaClO ₄	lgK:8.30 (3SF)	4	MPL	5398

Reaction No. 53226							
Sb+3 + 4<F-1> = Sb+3_F-1 (4)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:10.95 (4SF)	1	EES	
2	30	2	(H,Na)ClO ₄	lgK:10.95 (4SF)	5	MPL	13996
3	30	2	NaClO ₄	lgK:10.95 (4SF)	4	MPL	5398

Reaction No. 54251							
Sb+3_Cl-1 (3) + Cl-1 = Sb+3_Cl-1 (4)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	15	6.3	LiNO ₃	lgK:1.3 (2SF)	1	MDS	931

2	15	6.3	LiNO ₃	lgK:1.3(2SF)	1	MDS	14051
Note (2): Values given : 1.3 to 1.5							
3	25	0	Inf. Dilution	lgK:1.0(2SF)	1	EES	
4	25	4	H ₂ SO ₄	lgK:1.0(2SF)	1	MSL	931
5	25	4	H ₂ SO ₄	lgK:1.0(2SF)	1	MSL	14000

Reaction No. 54677							
0.5<Sb+3(2)_S-2(3)_s> + 1.5<S-2> = Sb+3_S-2(3)							
Note: The species S-2 does not exist in aqueous solution See reaction: H+1 + S-2 = H+1_S-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:0.89(2SF)	0	MSL	14015

Reaction No. 54678							
0.5<Sb2O3(s)> + 3<H+1> + 3<e-1> = Sb(s) + 1.5<H2O>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:8.1(2SF)	1	EDH	
2	25	0	H ₂ SO ₄	lgK:8.3(2SF)	2	EBP	23088
Note (2): EBP: 2<Sb+3_OH-1(4)> + 2<H+1> = Sb2O3(s) + 5<H2O>							
3	25	0	Inf. Dilution	E:0.15(2SF)	0	ENS	773
4	25	0	Inf. Dilution	E:0.08(0.05SD)	0	MEF	775
Note (4): Low wgt for inter-reaction consistency							
5	25	0	Inf. Dilution	E:0.152(3SF)	0	CRV	6330
Note (5): No account taken of changing ionic strength							
6	25	0	Inf. Dilution	E:0.1445(4SF)	2	CRV	13213
7	25	0.2:1.1	HClO ₄	E:0.152(3SF)	5	MEF	7829

Reaction No. 54679							
2<Sb+3(2)_S-2(3)_s> + H+1_S-2 + OH-1 = Sb+3(4)_S-2(7) + H ₂ O							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	3	Mixed salts	lgK:0.7(1SF)	1	MSL	14008
Note (1): Mean I calc. from exp. solutions (MF), Sb ₂ S ₃ + Na ₂ S, var. from 0.4 to 6.1							

Reaction No. 56880							
Co(s) + Sb(s) = CoSb(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dH:-10.(0.5SD)	5	ENS	802

Reaction No. 56881							
Co(s) + 2<Sb(s)> = CoSb2(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:8.755 (4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:8.700 (4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:6.514 (4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:5.201 (4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:4.327 (4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-11.94 (4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-13. (0.5SD)	4	ENS	802
8	25	0	Inf. Dilution	dH:-13. (0.5SD)	5	ENS	802
9	25	0	Inf. Dilution	dH:-12.0 (3SF)	4	MTD	6422
10	127	0	Inf. Dilution	dH:-12.0 (3SF)	4	MTD	6422
11	227	0	Inf. Dilution	dH:-12.0 (3SF)	4	MTD	6422
12	327	0	Inf. Dilution	dH:-11.9 (3SF)	4	MTD	6422

Reaction No. 56882							
Co(s) + 3<Sb(s)> = CoSb3(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:10.75 (4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:10.68 (4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:7.950 (4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:6.309 (4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:5.216 (4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-14.70 (4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-16. (0.5SD)	4	ENS	802
8	25	0	Inf. Dilution	dH:-16. (0.5SD)	5	ENS	802
9	25	0	Inf. Dilution	dH:-15.0 (3SF)	4	MTD	6422
10	127	0	Inf. Dilution	dH:-15.0 (3SF)	4	MTD	6422
11	227	0	Inf. Dilution	dH:-15.0 (3SF)	4	MTD	6422
12	327	0	Inf. Dilution	dH:-14.9 (3SF)	4	MTD	6422

Reaction No. 57450							
FeSbO4(s) + H2O + 3<H+1> = Fe+3 + Sb+5_OH-1(5)							
Note: Mineral name: tripuhyite							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

1	25	0	Inf. Dilution	lgK:-11.(2SF)	1	ELC	
2	25	0.195	HNO3	lgK:-10.68(0.10SD)	5	MSL	30052
Note (2): Authors also give deltaG of formation							

Reaction No. 57612							
Ni(s) + Sb(s) = NiSb(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:14.81(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:14.72(4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:11.07(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:8.874(4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:7.400(4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-20.20(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-15.6(0.1SD)	4	ENS	3006
8	25	0	Inf. Dilution	dH:-15.6(0.1SD)	5	ENS	3006
9	25	0	Inf. Dilution	dH:-20.0(3SF)	4	MTD	6422
10	127	0	Inf. Dilution	dH:-20.1(3SF)	4	MTD	6422
11	227	0	Inf. Dilution	dH:-20.2(3SF)	4	MTD	6422
12	327	0	Inf. Dilution	dH:-20.3(3SF)	4	MTD	6422

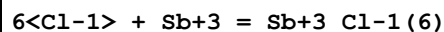
Reaction No. 58028							
Sb+3_OH-1(3) + OH-1 = Sb+3_OH-1(4)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:2.2(0.1SD)	3	CSM	8

Reaction No. 58290							
Sb+3_OH-1_DMSA-4 + 2<H2O> = Sb+3_OH-1(3) + H+1(2)_DMSA-4							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	NaClO4	lgK:-9.5(2SF)	0	MGL	1986
2	25	0	Inf. Dilution	lgK:-8.7(2SF)	1	ELC	

Reaction No. 58568							
6<F-1> + Sb+3 = Sb+3_F-1(6)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:36.(2SF)	1	EES	
2	25	0.5	KCl	lgK:36.(2SF)	3	MPL	140
3	25	0.5	KCl	lgK:36.(2SF)	4	MPL	14014

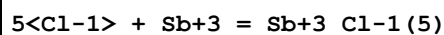
Note (3): Same value in Pavlov D Lazarov D Proc Acad Sci USSR 1958 118 19

Reaction No. 58756



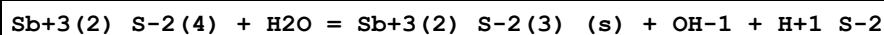
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:4.1(2SF)	1	EES	
2	25	4	HClO4	lgK:4.11(3SF)	2	MPL	140
3	25	4	HClO4	lgK:4.11(3SF)	2	MPL	13998

Reaction No. 58757



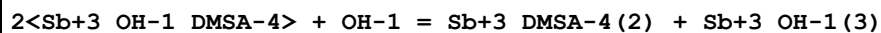
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	4.7	(H,Na)ClO4	lgK:1.10(3SF)	0	MPL	5296
Note (1): Grossly inconsistent - authors show differences without comment							
2	25	0	Inf. Dilution	lgK:4.7(2SF)	1	EES	
3	25	4	HClO4	lgK:4.72(3SF)	2	MPL	140
4	25	4	HClO4	lgK:4.72(3SF)	2	MPL	13998

Reaction No. 59325



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-1.9(2SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:1.310(4SF)	0	EOD	31708
3	30	0.5	Mixed salts	lgK:-1.16(3SF)	1	MSL	14009
Note (3): Mean I calc. from exp. solutions (MF), mixt. Sb2S3 + KCl + Na2S							
4	50	0	Inf. Dilution	lgK:1.306(4SF)	0	EOD	31708
5	100	0	Inf. Dilution	lgK:1.278(4SF)	0	EOD	31708
6	150	0	Inf. Dilution	lgK:1.244(4SF)	0	EOD	31708
7	200	0	Inf. Dilution	lgK:1.225(4SF)	0	EOD	31708
8	250	0	Inf. Dilution	lgK:1.126(4SF)	0	EOD	31708
9	300	0	Inf. Dilution	lgK:1.317(4SF)	0	EOD	31708
10	350	0	Inf. Dilution	lgK:1.534(4SF)	0	EOD	31708

Reaction No. 59419

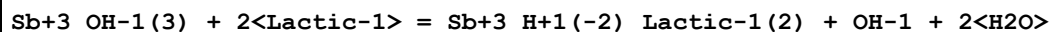


Note: Wrong ligand, DMSA-4 in original reaction (11023) is d1DMSA-4

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
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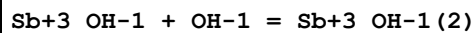
1	20	0.1	Unknown	lgK:-10.7 (3SF)	0	CRV	7271
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Reaction No. 60234



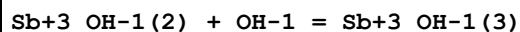
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0	Unknown	lgK:-5.08 (3SF)	5	MPH	23085
2	25	0	Inf. Dilution	lgK:-5.08 (3SF)	1	ESC	

Reaction No. 60589



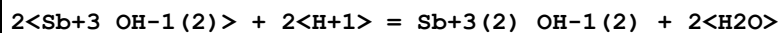
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:14.2 (3SF)	1	ELC	
2	25	5	Unknown	lgK:15.5 (3SF)	0	CRV	2556

Reaction No. 60590



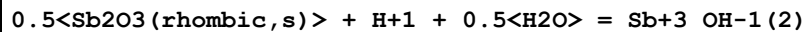
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:12.7 (3SF)	1	EOR	
2	25	0	Inf. Dilution	lgK:12.8 (3SF)	0	CRV	2556
3	25	3	Unknown	lgK:13.0 (3SF)	2	CRV	2556

Reaction No. 60591



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:0.8 (1SF)	1	EES	
2	25	5	(H,Na)ClO4	lgK:0.68 (2SF)	6	MSL	7780
3	25	5	Unknown	lgK:0.7 (1SF)	3	CRV	2556

Reaction No. 60592



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0:0.1	HCl	lgK:-3.11 (3SF)	4	MSL	8047
2	25	0	Inf. Dilution	lgK:-3.11 (3SF)	0	MSL	140
3	25	0	Inf. Dilution	lgK:-3.09 (3SF)	3	CRV	2556
4	25	0.2:1.1	HClO4	lgK:-3.1 (2SF)	3	MSL	7829
5	25	5	(H,Na)ClO4	lgK:-3.06 (3SF)	6	MSL	7780
6	25	5	Unknown	lgK:-3.06 (3SF)	3	CRV	2556

Reaction No. 60593							
$0.5\text{Sb}_2\text{O}_3(\text{cubic},\text{s}) + \text{H}^+ + 0.5\text{H}_2\text{O} = \text{Sb}^{3+}_{\text{OH}-1(2)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: -3.18(3\text{SF})$	1	ELC	
2	25	5	(H,Na)ClO ₄	$\lg K: -3.18(3\text{SF})$	6	MSL	7780
3	25	5	Unknown	$\lg K: -3.18(3\text{SF})$	3	CRV	2556

Reaction No. 61560							
$\text{Sb}^{3+} + \text{H}^+_{\text{Lys}-1} = \text{Sb}^{3+}_{\text{H}^+_{\text{Lys}-1}}$							
Note: Sb ³⁺ is not what the authors had in solution; does it exist?							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	15	1	KNO ₃	$\lg K: 3.08(3\text{SF})$	0	MGL	6000
2	15	1	KNO ₃	$\lg K: 3.08(3\text{SF})$	0	MGL	8989
3	30	1	KNO ₃	$\lg K: 2.35(3\text{SF})$	0	MGL	6000
4	30	1	KNO ₃	$\lg K: 2.35(3\text{SF})$	0	MGL	8989
5	15	1	KNO ₃	$\text{dH}: 2.13(3\text{SF})$	0	MCL	6000

Reaction No. 61561							
$\text{Sb}^{3+}_{\text{H}^+_{\text{Lys}-1}} + \text{H}^+_{\text{Lys}-1} = \text{Sb}^{3+}_{\text{H}^+_{(2)}_{\text{Lys}-1(2)}}$							
Note: Stoichiometry of species unknown, linked to 61560							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	15	1	KNO ₃	$\lg K: 2.86(3\text{SF})$	0	MGL	6000
2	15	1	KNO ₃	$\lg K: 2.86(3\text{SF})$	0	MGL	8989
3	30	1	KNO ₃	$\lg K: 3.20(3\text{SF})$	0	MGL	6000
4	30	1	KNO ₃	$\lg K: 3.20(3\text{SF})$	0	MGL	8989

Reaction No. 66119							
$\text{Sb}^{3+}_{\text{OH}-1(3)} + \text{H}^+_{\text{HOEDTA}-3} = \text{Sb}^{3+}_{\text{OH}-1_{\text{HOEDTA}-3}} + \text{OH}^- + \text{H}_2\text{O}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	NaClO ₄	$\lg K: -4.58(2\text{SF})$	6	MGL	2038
2	20	0.1	Unknown	$\lg K: 4.58(3\text{SF})$	0	CRV	552
Note (2): Sign is wrong							
3	25	0	Inf. Dilution	$\lg K: -4.5(2\text{SF})$	1	ESC	

Reaction No. 66127							
$\text{Sb}^{3+}_{\text{OH}-1_{\text{EDTA}-4}} + \text{OH}^- = \text{Sb}^{3+}_{\text{OH}-1(2)_{\text{EDTA}-4}}$							

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:5.9(2SF)	1	ELC	
2	25	0.1	Unknown	lgK:7.46(3SF)	0	CRV	552
Note (2): Value from Ozer Bogucki 71. See Sb+3_OH-1(2)_EDTA-1 + H+1							

Reaction No. 66433							
2<Sb(s)> + 3<Se(s)> = Sb+3(2)_Se-2(3)_(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:22.06(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:21.92(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:16.36(4SF)	3	ENS	6422
4	227	0	Inf. Dilution	lgK:12.97(4SF)	2	ENS	6422
5	327	0	Inf. Dilution	lgK:10.39(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-30.1(3SF)	4	EFE	6422
7	25	0	Inf. Dilution	dG:-125.6(4SF)kJ	3	CRV	30961
8	25	0	Inf. Dilution	dH:-30.5(3SF)	4	MTD	6422
9	25	0	Inf. Dilution	dH:-127.6(4SF)kJ	3	CRV	30961
10	127	0	Inf. Dilution	dH:-30.6(3SF)	1	MTD	6422
11	227	0	Inf. Dilution	dH:-35.1(3SF)	0	MTD	6422
12	327	0	Inf. Dilution	dH:-35.8(3SF)	0	MTD	6422
13	25	0	Inf. Dilution	Cp:124.98(5SF)J/K	3	CRV	30961

Reaction No. 66434							
2<Sb(s)> + 3<S(s)> + 6<O2(g)> = Sb+3(2)_SO4-2(3)_(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:362.0(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:359.4(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:254.8(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:191.8(4SF)	3	ENS	6422
5	327	0	Inf. Dilution	lgK:149.7(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-493.9(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-574.2(4SF)	4	MTD	6422
8	25	0	Inf. Dilution	dH:-574.2(4SF)	4	CRV	7382
9	127	0	Inf. Dilution	dH:-576.2(4SF)	4	MTD	6422
10	227	0	Inf. Dilution	dH:-577.4(4SF)	4	MTD	6422
11	327	0	Inf. Dilution	dH:-578.0(4SF)	4	MTD	6422

Reaction No. 66435							
$3\text{S(s)} + 2\text{Sb(s)} = \text{Sb}_3\text{(2)}_{\text{S-2(3)}}_{\text{(s)}}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:30.7 (3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:24.58 (4SF)	0	ENS	6422
3	25	0	Inf. Dilution	lgK:26.25 (4SF)	0	ENS	6494
4	27	0	Inf. Dilution	lgK:24.43 (4SF)	0	ENS	6422
5	27	0	Inf. Dilution	lgK:26.09 (4SF)	0	ENS	6494
6	127	0	Inf. Dilution	lgK:18.22 (4SF)	0	ENS	6422
7	127	0	Inf. Dilution	lgK:19.46 (4SF)	0	ENS	6494
8	227	0	Inf. Dilution	lgK:14.30 (4SF)	0	ENS	6422
9	227	0	Inf. Dilution	lgK:15.28 (4SF)	0	ENS	6494
10	327	0	Inf. Dilution	lgK:11.61 (4SF)	0	ENS	6422
11	327	0	Inf. Dilution	lgK:12.42 (4SF)	0	ENS	6494
12	25	0	Inf. Dilution	dG:-32. (2SF)	0	EFE	600
13	25	0	Inf. Dilution	dG:-33.53 (4SF)	0	EFE	6422
14	25	0	Inf. Dilution	dG:-35.80 (4SF)	0	EFE	6494
15	25	0	Inf. Dilution	dG:-41.5 (3SF)	0	CRV	7382
16	25	0	Inf. Dilution	dG:-173.6 (4SF) kJ	0	EFE	9015
17	25	0	Inf. Dilution	dG:-41.49 (4SF)	0	EFE	23127
Note (17): Copied from Wagman1982 [9015]							
18	25	0	Inf. Dilution	dG:-173.60 (5SF) kJ	0	EOD	29489
19	25	0	Inf. Dilution	dG:-41.5 (3SF)	0	EFE	31675
20	25	0	Inf. Dilution	dG:-34.156 (5SF)	0	EOD	31708
21	25	0	Inf. Dilution	dH:-33.9 (3SF)	2	MTD	6422
22	25	0	Inf. Dilution	dH:-36.19 (4SF)	0	MTD	6494
23	25	0	Inf. Dilution	dH:-41.8 (3SF)	0	CRV	7382
24	25	0	Inf. Dilution	dH:-174.90 (5SF) kJ	0	EOD	29489
25	127	0	Inf. Dilution	dH:-35.5 (3SF)	0	MTD	6422
26	127	0	Inf. Dilution	dH:-37.79 (4SF)	0	MTD	6494
27	227	0	Inf. Dilution	dH:-36.5 (3SF)	0	MTD	6422
28	227	0	Inf. Dilution	dH:-38.89 (4SF)	0	MTD	6494
29	327	0	Inf. Dilution	dH:-37.3 (3SF)	0	MTD	6422
30	327	0	Inf. Dilution	dH:-39.67 (4SF)	0	MTD	6494

Reaction No. 66436

0.5<Cl ₂ (g)> + 0.5<O ₂ (g)> + Sb(s) = SbOCl(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:57.59(4SF)	3	ENS	6422
2	27	0	Inf. Dilution	lgK:57.18(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:40.94(4SF)	2	ENS	6422
4	227	0	Inf. Dilution	lgK:31.24(4SF)	1	ENS	6422
5	327	0	Inf. Dilution	lgK:24.80(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-78.56(4SF)	3	EFE	6422
Note (6): Only existing value; no experimental value found							
7	25	0	Inf. Dilution	dH:-89.4(3SF)	2	MTD	6422
8	25	0	Inf. Dilution	dH:-89.4(3SF)	3	CRV	7382
9	127	0	Inf. Dilution	dH:-89.0(3SF)	1	MTD	6422
10	227	0	Inf. Dilution	dH:-88.5(3SF)	0	MTD	6422
11	327	0	Inf. Dilution	dH:-88.1(3SF)	0	MTD	6422

Reaction No. 66437							
2.5<O ₂ (g)> + 2<Sb(s)> = Sb ₂ O ₅ (s)							
Note: Sb ₂ O ₅ (s) very probably not present in original experiment							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:145.3(4SF)	0	ENS	6422
2	27	0	Inf. Dilution	lgK:144.2(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:101.9(4SF)	0	ENS	6422
4	227	0	Inf. Dilution	lgK:76.51(4SF)	0	ENS	6422
5	327	0	Inf. Dilution	lgK:59.59(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-200.5(4SF)	2	EFE	600
Note (6): Copied from NBS1952 [15886]							
7	25	0	Inf. Dilution	dG:-198.2(4SF)	2	EFE	6422
Note (7): Copied from Wagman1982 [9015]							
8	25	0	Inf. Dilution	dG:-198.2(4SF)	2	CRV	7382
9	25	0	Inf. Dilution	dG:-829.2(4SF) kJ	3	EFE	9015
10	25	0	Inf. Dilution	dG:-200.5(4SF)	2	EFE	13986
Note (10): Copied from NBS1952 [15886]							
11	25	0	Inf. Dilution	dG:-200.5(4SF)	2	EFE	15886
12	25	0	Inf. Dilution	dG:-198.2(4SF)	3	EFE	23127
Note (12): Copied from Wagman1982 [9015]							
13	25	0	Inf. Dilution	dG:-829.20(5SF) kJ	2	EOD	29489

Note (13): Copied from Wagman1982 [9015]							
14	25	0	Inf. Dilution	dG:-198.19 (5SF)	2	CRV	31291
15	25	0	Inf. Dilution	dG:-198.2 (4SF)	3	EFE	31675
16	25	0	Inf. Dilution	dH:-232.3 (4SF)	3	MTD	6422
17	25	0	Inf. Dilution	dH:-232.3 (4SF)	3	CRV	7382
18	25	0	Inf. Dilution	dH:-971.90 (5SF) kJ	2	EOD	29489
19	127	0	Inf. Dilution	dH:-232.3 (4SF)	1	MTD	6422
20	227	0	Inf. Dilution	dH:-232.3 (4SF)	0	MTD	6422
21	327	0	Inf. Dilution	dH:-232.3 (4SF)	0	MTD	6422

Reaction No. 66438							
$2\text{Sb(s)} + 2\text{O}_2\text{(g)} = \text{Sb}_2\text{O}_4\text{(s)}$							
Note: Sb ₂ O ₄ (s) in doubt in the original experiment							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:139.4 (4SF)	2	ENS	6422
2	27	0	Inf. Dilution	lgK:138.5 (4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:98.97 (4SF)	0	ENS	6422
4	227	0	Inf. Dilution	lgK:75.31 (4SF)	0	ENS	6422
5	327	0	Inf. Dilution	lgK:59.56 (4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-165.9 (4SF)	0	EFE	600
7	25	0	Inf. Dilution	dG:-190.2 (4SF)	2	EFE	6422
Note (7): Copied from Wagman1982 [9015]							
8	25	0	Inf. Dilution	dG:-190.2 (4SF)	2	CRV	7382
9	25	0	Inf. Dilution	dG:-795.7 (4SF) kJ	2	EFE	9015
10	25	0	Inf. Dilution	dG:-165.9 (4SF)	0	EFE	13986
11	25	0	Inf. Dilution	dG:-188. (3SF)	0	EFE	15886
12	25	0	Inf. Dilution	dG:-190.18 (5SF)	2	EFE	23127
Note (12): Copied from Wagman1982 [9015]							
13	25	0	Inf. Dilution	dG:-795.70 (5SF) kJ	2	EOD	29489
Note (13): Copied from Wagman1982 [9015]							
14	25	0	Inf. Dilution	dG:-190.24 (5SF)	2	CRV	31291
15	25	0	Inf. Dilution	dG:-190.2 (4SF)	2	EFE	31675
16	25	0	Inf. Dilution	dH:-216.9 (4SF)	4	MTD	6422
17	25	0	Inf. Dilution	dH:-216.9 (4SF)	4	CRV	7382
18	25	0	Inf. Dilution	dH:-907.50 (5SF) kJ	2	EOD	29489
19	127	0	Inf. Dilution	dH:-216.7 (4SF)	1	MTD	6422

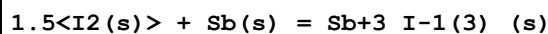
20	227	0	Inf. Dilution	dH:-216.4 (4SF)	0	MTD	6422
21	327	0	Inf. Dilution	dH:-216.1 (4SF)	0	MTD	6422

Reaction No. 66439							
1.5<O2(g)> + 2<Sb(s)> = Sb2O3(rhombic,s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:109.1 (4SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:109.7 (4SF)	0	ENS	6422
3	27	0	Inf. Dilution	lgK:109.0 (4SF)	0	ENS	6422
4	127	0	Inf. Dilution	lgK:78.14 (4SF)	0	ENS	6422
5	227	0	Inf. Dilution	lgK:59.68 (4SF)	0	ENS	6422
6	327	0	Inf. Dilution	lgK:47.39 (4SF)	0	ENS	6422
7	25	0	Inf. Dilution	dG:-149.7 (4SF)	0	EFE	6422
8	25	0	Inf. Dilution	dG:-149. (3SF)	2	EFE	13986
9	25	0	Inf. Dilution	dG:-625.9 (2.1SD) kJ	0	EOD	29879
10	25	0	Inf. Dilution	dG:-148.190 (6SF)	4	EOD	31708
11	25	0	Inf. Dilution	dH:-169.3 (4SF)	0	MTD	6422
12	127	0	Inf. Dilution	dH:-169.1 (4SF)	0	MTD	6422
13	227	0	Inf. Dilution	dH:-168.8 (4SF)	0	MTD	6422
14	327	0	Inf. Dilution	dH:-168.5 (4SF)	0	MTD	6422

Reaction No. 66440							
1.5<O2(g)> + 2<Sb(s)> = Sb2O3(cubic,s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:111.9 (4SF)	0	ELC	
2	25	0	Inf. Dilution	lgK:111.1 (4SF)	0	ENS	6422
3	27	0	Inf. Dilution	lgK:110.4 (4SF)	0	ENS	6422
4	127	0	Inf. Dilution	lgK:79.04 (4SF)	0	ENS	6422
5	227	0	Inf. Dilution	lgK:60.31 (4SF)	0	ENS	6422
6	327	0	Inf. Dilution	lgK:47.86 (4SF)	0	ENS	6422
7	25	0	Inf. Dilution	dG:-151.6 (4SF)	0	EFE	6422
8	25	0	Inf. Dilution	dG:-147. (3SF)	2	EFE	13986
9	25	0	Inf. Dilution	dG:-633.2 (2.1SD) kJ	0	EOD	29879
10	25	0	Inf. Dilution	dG:-149.970 (6SF)	4	EOD	31708
11	25	0	Inf. Dilution	dH:-172.2 (4SF)	0	MTD	6422
12	127	0	Inf. Dilution	dH:-171.7 (4SF)	0	MTD	6422

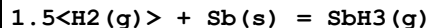
13	227	0	Inf. Dilution	dH:-171.1 (4SF)	0	MTD	6422
14	327	0	Inf. Dilution	dH:-170.5 (4SF)	0	MTD	6422

Reaction No. 66441



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:17.55 (4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:17.44 (4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:12.94 (4SF)	2	ENS	6422
4	171	0	Inf. Dilution	lgK:11.31 (4SF)	2	ENS	6422
5	25	0	Inf. Dilution	dG:-23.94 (4SF)	4	EFE	6422
6	25	0	Inf. Dilution	dG:-23.941 (5SF)	3	CRV	31291
7	25	0	Inf. Dilution	dH:-24.0 (3SF)	4	MTD	6422
8	25	0	Inf. Dilution	dH:-24.0 (3SF)	4	CRV	7382
9	25	0	Inf. Dilution	dH:-24.000 (5SF)	3	CRV	31291
10	127	0	Inf. Dilution	dH:-29.9 (3SF)	1	MTD	6422
11	171	0	Inf. Dilution	dH:-30.3 (3SF)	0	MTD	6422

Reaction No. 66442



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-25.86 (4SF)	2	ENS	6422
2	27	0	Inf. Dilution	lgK:-25.71 (4SF)	1	ENS	6422
3	127	0	Inf. Dilution	lgK:-19.44 (4SF)	1	ENS	6422
4	227	0	Inf. Dilution	lgK:-15.74 (4SF)	0	ENS	6422
5	327	0	Inf. Dilution	lgK:-13.29 (4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:35.3 (3SF)	2	EFE	600
7	25	0	Inf. Dilution	dG:35.28 (4SF)	2	EFE	6422
8	25	0	Inf. Dilution	dG:147.75 (5SF) kJ	3	EFE	7275
9	25	0	Inf. Dilution	dG:35.31 (4SF)	2	CRV	7382
10	25	0	Inf. Dilution	dG:145.110 (6SF) kJ	3	EFE	9015
11	25	0	Inf. Dilution	dG:33.31 (4SF)	3	EFE	31708
12	25	0	Inf. Dilution	dH:34.68 (4SF)	4	MTD	6422
13	25	0	Inf. Dilution	dH:145.11 (5SF) kJ	5	EFE	7275
14	25	0	Inf. Dilution	dH:34.681 (5SF)	4	CRV	7382
15	27	0	Inf. Dilution	dH:34.67 (4SF)	4	MTD	6422

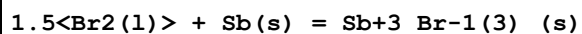
16	127	0	Inf. Dilution	dH:34.11 (4SF)	4	MTD	6422
17	227	0	Inf. Dilution	dH:33.69 (4SF)	4	MTD	6422
18	327	0	Inf. Dilution	dH:33.35 (4SF)	4	MTD	6422

Reaction No. 66443							
Sb(s) + 1.5<F2(g)> = Sb+3_F-1(3)_(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:148.8 (4SF)	2	ENS	6422
2	27	0	Inf. Dilution	lgK:147.8 (4SF)	2	ENS	6422
3	127	0	Inf. Dilution	lgK:108.0 (4SF)	0	ENS	6422
4	227	0	Inf. Dilution	lgK:84.14 (4SF)	0	ENS	6422
5	287	0	Inf. Dilution	lgK:73.96 (4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-199.8 (4SF)	1	EOD	600
7	25	0	Inf. Dilution	dG:-202.9 (4SF)	2	EFE	6422
8	25	0	Inf. Dilution	dH:-218.8 (4SF)	3	MTD	6422
9	25	0	Inf. Dilution	dH:-218.8 (4SF)	3	CRV	7382
10	127	0	Inf. Dilution	dH:-218.3 (4SF)	0	MTD	6422
11	227	0	Inf. Dilution	dH:-217.7 (4SF)	0	MTD	6422
12	287	0	Inf. Dilution	dH:-217.2 (4SF)	0	MTD	6422

Reaction No. 66444							
1.5<Cl2(g)> + Sb(s) = Sb+3_Cl-1(3)_(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:56.49 (4SF)	0	ENS	6422
2	27	0	Inf. Dilution	lgK:56.08 (4SF)	0	ENS	6422
3	73	0	Inf. Dilution	lgK:47.28 (4SF)	0	ENS	6422
4	25	0	Inf. Dilution	dG:-77.62 (4SF)	0	EFE	600
5	25	0	Inf. Dilution	dG:-323.67 (5SF) kJ	2	CRV	6330
6	25	0	Inf. Dilution	dG:-77.07 (4SF)	0	EFE	6422
7	25	0	Inf. Dilution	dG:-77.37 (4SF)	0	CRV	7382
8	25	0	Inf. Dilution	dG:-323.67 (5SF) kJ	2	EFE	9015
9	25	0	Inf. Dilution	dG:-77.62 (4SF) kJ	0	EFE	15886
10	25	0	Inf. Dilution	dG:-323.67 (5SF) kJ	2	EOD	29489
Note (10): Copied from Wagman1982 [9015]							
11	25	0	Inf. Dilution	dG:-77.37 (4SF)	0	EFE	31675
12	25	0	Inf. Dilution	dH:-382.2 (4SF) kJ	1	CRV	6330

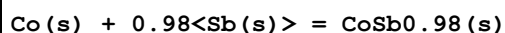
13	25	0	Inf. Dilution	dH:-91.1 (3SF)	1	MTD	6422
14	25	0	Inf. Dilution	dH:-91.34 (4SF)	1	CRV	7382
15	25	0	Inf. Dilution	dH:-382.17 (5SF) kJ	1	EOD	29489
16	73	0	Inf. Dilution	dH:-90.7 (3SF)	0	MTD	6422

Reaction No. 66445



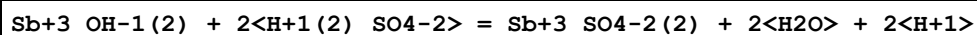
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:42.12 (4SF)	0	ENS	6422
2	27	0	Inf. Dilution	lgK:41.83 (4SF)	0	ENS	6422
3	95	0	Inf. Dilution	lgK:32.81 (4SF)	2	ENS	6422
4	25	0	Inf. Dilution	dG:-239.3 (4SF) kJ	0	CRV	6330
5	25	0	Inf. Dilution	dG:-57.45 (4SF)	3	EFE	6422
6	25	0	Inf. Dilution	dG:-57.2 (3SF)	3	CRV	7382
7	25	0	Inf. Dilution	dG:-239.3 (4SF) kJ	2	EOD	29489
8	25	0	Inf. Dilution	dH:-259.4 (4SF) kJ	3	CRV	6330
9	25	0	Inf. Dilution	dH:-62.0 (3SF)	3	MTD	6422
10	25	0	Inf. Dilution	dH:-62.0 (3SF)	3	CRV	7382
11	25	0	Inf. Dilution	dH:-259.4 (4SF) kJ	2	EOD	29489
12	95	0	Inf. Dilution	dH:-72.5 (3SF)	0	MTD	6422

Reaction No. 66648



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:7.124 (4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:7.079 (4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:5.259 (4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:4.169 (4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:3.444 (4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-9.72 (3SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-10.0 (3SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-10.0 (3SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-10.0 (3SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-9.9 (2SF)	4	MTD	6422

Reaction No. 66854



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	18	14:18	H2SO4	lgK:-1.0(2SF)	4	MSP	13999
Note (1): Temperature guess (MF)							

Reaction No. 66855							
Sb+3_OH-1(2) + SO4-2 = Sb+3_OH-1(2)_SO4-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	18	1:4	H2SO4	lgK:0.3(1SF)	2	MSP	13999
Note (1): Temperature guess (MF)							
2	25	0	Inf. Dilution	lgK:0.65(2SF)	1	EAE	

Reaction No. 66856							
Sb+3(2)_S-2(3)_(s) = 2<Sb+3> + 3<S-2>							
Note: The species S-2 does not exist in aqueous solution See reaction: H+1 + S-2 = H+1_S-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-96.4(3SF)	0	ELC	
2	25	0	Inf. Dilution	lgK:-92.77(4SF)	0	MSL	140
3	25	0	Inf. Dilution	lgK:-93.(2SF)	0	ENS	773
4	25	0	Inf. Dilution	lgK:-92.8(3SF)	0	MSL	14015
Note (4): lgK depends on B11 which is unreliable							

Reaction No. 67228							
2<Sb(s)> + 3<Te(s)> = Sb+3(2)_Te-2(3)_(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:10.26(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:10.20(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:7.737(4SF)	2	ENS	6422
4	227	0	Inf. Dilution	lgK:6.258(4SF)	1	ENS	6422
5	327	0	Inf. Dilution	lgK:5.265(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-13.99(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-13.5(3SF)	3	MTD	6422
8	127	0	Inf. Dilution	dH:-13.5(3SF)	1	MTD	6422
9	227	0	Inf. Dilution	dH:-13.6(3SF)	0	MTD	6422
10	327	0	Inf. Dilution	dH:-13.7(3SF)	0	MTD	6422

Reaction No. 67353							
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In(s) + Sb(s) = InSb(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:4.448(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:4.415(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:3.083(4SF)	2	ENS	6422
4	227	0	Inf. Dilution	lgK:2.221(4SF)	1	ENS	6422
5	327	0	Inf. Dilution	lgK:1.621(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-6.068(4SF)	3	EFE	6422
7	25	0	Inf. Dilution	dG:-6.1(2SF)	4	CRV	12011
8	25	0	Inf. Dilution	dH:-7.29(3SF)	3	MTD	6422
9	25	0	Inf. Dilution	dH:-7.3(2SF)	4	CRV	12011
10	127	0	Inf. Dilution	dH:-7.35(3SF)	1	MTD	6422
11	227	0	Inf. Dilution	dH:-8.20(3SF)	0	MTD	6422
12	327	0	Inf. Dilution	dH:-8.26(3SF)	0	MTD	6422

Reaction No. 67394							
Fe(s) + 4<S(s)> + 2<Sb(s)> = FeSb2S4(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:44.81(4SF)	0	ENS	6494
2	27	0	Inf. Dilution	lgK:44.54(4SF)	0	ENS	6494
3	127	0	Inf. Dilution	lgK:33.36(4SF)	0	ENS	6494
4	227	0	Inf. Dilution	lgK:26.40(4SF)	0	ENS	6494
5	327	0	Inf. Dilution	lgK:21.69(4SF)	0	ENS	6494
6	25	0	Inf. Dilution	dG:-61.14(4SF)	0	EFE	6494
7	25	0	Inf. Dilution	dH:-61.23(4SF)	0	MTD	6494
8	127	0	Inf. Dilution	dH:-63.12(4SF)	0	MTD	6494
9	227	0	Inf. Dilution	dH:-64.32(4SF)	0	MTD	6494
10	327	0	Inf. Dilution	dH:-65.22(4SF)	0	MTD	6494

Reaction No. 70520							
Sb+3(2)_S-2(3)_(s) + S-2 = Sb+3(2)_S-2(4)							
Note: The species S-2 does not exist in aqueous solution See reaction: H+1 + S-2 = H+1_S-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:6.2(2SF)	0	ELC	
2	30	0.5	Mixed salts	lgK:2.1(2SF)	0	MSL	14009
Note (2): Mean I calc. from exp. solutions (MF), mixt. Sb2S3 + KCl + Na2S							

Reaction No. 70543							
$\text{Sb406(cubic,s)} + 4\text{<H+1>} + 2\text{<H2O>} = 4\text{<Sb+3_OH-1(2)>}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-16.4(3SF)	1	ELC	
2	25	5	NaClO4	lgK:-12.7(3SF)	0	MSL	5398

Reaction No. 70544							
$\text{Sb406(rhombic,s)} + 4\text{<H+1>} + 2\text{<H2O>} = 4\text{<Sb+3_OH-1(2)>}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-12.7(3SF)	1	ELC	
2	25	5	NaClO4	lgK:-12.2(3SF)	3	MSL	5398

Reaction No. 70545							
$\text{Sb406(cubic,s)} + 8\text{<H+1>} = 2\text{<Sb+3(2)_OH-1(2)>} + 2\text{<H2O>}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-14.8(3SF)	1	ELC	
2	25	5	NaClO4	lgK:-11.71(4SF)	3	MSL	5398

Reaction No. 70546							
$1.5\text{<H2(g)>} + 1.5\text{<O2(g)>} + \text{Sb(s)} = \text{Sb+3_OH-1(3)_(s)}$							
Note: Sb+3_OH-1(3)_(s) as such never proved to exist							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:112.7(4SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-163.8(4SF)	0	CRV	7382
3	25	0	Inf. Dilution	dG:-685.2(4SF) kJ	0	EFE	9015
4	25	0	Inf. Dilution	dG:-685.20(5SF) kJ	0	EOD	29489
Note (4): Copied from Wagman1982 [9015]							
5	25	0	Inf. Dilution	dG:-163.8(4SF)	0	EFE	31675

Reaction No. 70547							
$1.5\text{<H2(g)>} + 1.5\text{<O2(g)>} + \text{Sb(s)} = \text{Sb+3_OH-1(3)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:112.4(4SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-154.1(4SF)	0	CRV	7382
Note (2): Ref. dupl. data under $0.5\text{<H2(g)>} + \text{O2(g)} + \text{Sb(s)} = \text{H+1_SbO2-1}$							
3	25	0	Inf. Dilution	dG:-644.7(4SF) kJ	3	EFE	9015

4	25	0	Inf. Dilution	dG:-154.1(4SF)	0	CRV	9029
Note (4): lgK = 113.0							
5	25	0	Inf. Dilution	dG:-154.1(4SF)	4	CRV	12011
6	25	0	Inf. Dilution	dG:-644.7(4SF) kJ	2	EOD	29489
Note (6): Copied from Wagman1982 [9015]							
7	25	0	Inf. Dilution	dG:-644.4(1.1SD) kJ	2	EOD	29879
8	25	0	Inf. Dilution	dG:-154.1(4SF)	2	EFE	31675
9	25	0	Inf. Dilution	dH:-184.9(4SF)	3	CRV	7382
Note (9): Ref. dupl. data under 0.5<H2(g)> + O2(g) + Sb(s) = H+1_SbO2-1							
10	25	0	Inf. Dilution	dH:-184.9(4SF)	3	CRV	9029
11	25	0	Inf. Dilution	dH:-184.9(4SF)	4	CRV	12011
12	25	0	Inf. Dilution	dH:-773.6(4SF) kJ	2	EOD	29489

Reaction No. 70548							
1.5<Cl2(g)> + Sb(s) = Sb+3_Cl-1(3)_ (g)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-72.3(3SF)	2	EFE	600
2	25	0	Inf. Dilution	dG:-72.0(3SF)	2	CRV	7382
3	25	0	Inf. Dilution	dG:-301.2(4SF) kJ	3	EFE	9015
4	25	0	Inf. Dilution	dG:-72.3(3SF)	2	EFE	15886
5	25	0	Inf. Dilution	dG:-72.0(3SF)	3	EFE	31675
6	25	0	Inf. Dilution	dH:-75.0(3SF)	4	CRV	7382

Reaction No. 70549							
2.5<Cl2(g)> + Sb(s) = SbCl5(l)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-83.7(3SF)	2	CRV	7382
2	25	0	Inf. Dilution	dG:-350.1(4SF) kJ	3	EFE	9015
3	25	0	Inf. Dilution	dG:-83.7(3SF)	3	EFE	31675
4	25	0	Inf. Dilution	dH:-105.2(4SF)	2	CRV	7382

Reaction No. 71007							
Sb4O6(rhombic,s) + 8<H+1> = 2<Sb+3(2)_OH-1(2)> + 2<H2O>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-11.1(3SF)	1	ELC	
2	25	5	NaClO4	lgK:-10.86(4SF)	3	MSL	5398

Reaction No. 71008							
$\text{Sb}+3_{\text{OH}-1(3)} + 3\text{H}+1 = \text{Sb}+3 + 3\text{H}_2\text{O}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: -0.73(2\text{SF})$	5	CRV	31301
2	25	0	Inf. Dilution	$\lg K: -0.73(2\text{SF})$	5	CRV	31495

Reaction No. 71009							
$\text{Sb}_{205}(\text{s}) + 4\text{H}+1 + 4\text{e}-1 = \text{Sb}_{203}(\text{rhombic}, \text{s}) + 2\text{H}_2\text{O}$							
Note: $\text{Sb}_{205}(\text{s})$ very probably not present in original experiment							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: 46.9(3\text{SF})$	1	ELC	
2	25	0	Inf. Dilution	$E: 0.649(3\text{SF})$	0	CRV	6330
3	25	0	Inf. Dilution	$E: 0.649(3\text{SF})$	0	EMU	13986

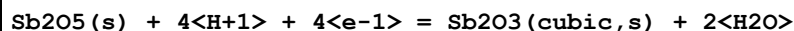
Reaction No. 71010							
$\text{Sb}_{205}(\text{s}) + 4\text{H}+1 + 4\text{e}-1 = \text{Sb}_{203}(\text{s}) + 2\text{H}_2\text{O}$							
Note: $\text{Sb}_{205}(\text{s})$ very probably not present in original experiment							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: 46.1(3\text{SF})$	1	ELC	
2	25	0	Inf. Dilution	$E: 0.692(3\text{SF})$	0	EFE	600
3	25	0	Inf. Dilution	$E: 0.69(2\text{SF})$	0	ENS	773
4	25	0	Inf. Dilution	$E: 0.8(0.03\text{SD})$	0	MDG	775
Note (4): Low wgt for inter-reaction consistency							
5	25	0	Inf. Dilution	$E: 0.649(3\text{SF})$	0	CRV	6330
Note (5): This is Pitman's "rhombic" value							
6	25	0	Inf. Dilution	$E: 0.671(3\text{SF})$	0	CRV	6330
Note (6): This is Pitman's "cubic" value							

Reaction No. 71011							
$0.5\text{Sb}+3(2)_{\text{S}-2(3)}(\text{s}) + 3\text{OH}-1 = 0.5\text{Sb}+3_{\text{S}-2(3)} + 0.5\text{Sb}+3_{\text{OH}-1(6)}$							
Note: $\text{Sb}+3_{\text{OH}-1(6)}$ is considered not to exist							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: 4.31(3\text{SF})$	0	MSL	14015

Reaction No. 71012							
$\text{Sb}(\text{s}) + 3\text{H}+1 + 3\text{e}-1 = \text{SbH}_3(\text{s})$							

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:-0.510 (3SF)	5	EFE	600
2	25	0	Inf. Dilution	E:-0.51 (2SF)	3	ENS	773
3	25	0	Inf. Dilution	E:-0.51 (0.01SD)	0	MDG	775
4	25	0	Inf. Dilution	E:-0.510 (3SF)	3	CRV	6330

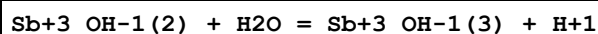
Reaction No. 71392



Note: Sb₂O₅(s) very probably not present in original experiment

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:47.1 (3SF)	1	ELC	
2	25	0	Inf. Dilution	E:0.671 (3SF)	0	CRV	6330
3	25	0	Inf. Dilution	E:0.671 (3SF)	0	EFE	13986

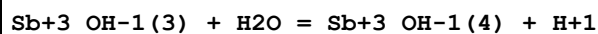
Reaction No. 72343



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	22	0	Inf. Dilution	lgK:-1.385 (4SF)	4	MSP	21754
2	23	0.02:0.1	HClO ₄	lgK:-1.42 (3SF)	3	MSP	14004
3	25	0	Inf. Dilution	lgK:-0.87 (2SF)	0	MSL	140
4	25	0	Inf. Dilution	lgK:-1.41 (3SF)	5	EOD	403
5	25	0	Inf. Dilution	lgK:-0.87 (0.02SD)	0	ENS	5559
6	25	0	Inf. Dilution	lgK:-1.41 (0.05SD)	4	CRV	6478
7	25	0	Inf. Dilution	lgK:-0.87 (2SF)	0	EFE	13986
8	25	0	Inf. Dilution	lgK:-1.4 (2SF)	4	MSL	19879
9	25	0	Inf. Dilution	lgK:-1.30 (3SF)	2	CRV	31301
10	25	0	Inf. Dilution	lgK:-1.30 (3SF)	3	CRV	31495
11	25	1	(H,Na)ClO ₄	lgK:-1.05 (3SF)	5	MSP	13993
12	25	3	(H,Na)ClO ₄	lgK:-1.23 (0.06SD)	5	MDS	13992
13	25	3	NaClO ₄	lgK:-1.24 (3SF)	3	MSL	5398
14	50	0	Inf. Dilution	lgK:-1.236 (4SF)	4	MSP	21754
15	75	0	Inf. Dilution	lgK:-1.078 (4SF)	4	MSP	21754
16	100	0	Inf. Dilution	lgK:-0.94 (2SF)	4	MSP	21754
17	22	0	Inf. Dilution	dH:-3.674 (4SF)	4	MTD	21754
18	25	0	Inf. Dilution	dH:5.1 (0.6SD)	4	CRV	6478
19	50	0	Inf. Dilution	dH:-2.897 (4SF)	4	MTD	21754

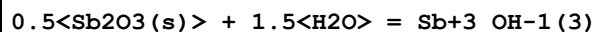
20	75	0	Inf. Dilution	dH:-2.04 (3SF)	4	MTD	21754
21	100	0	Inf. Dilution	dH:-1.02 (3SF)	4	MTD	21754
22	22	0	Inf. Dilution	dS:-6.12 (3SF)	4	EFE	21754
23	50	0	Inf. Dilution	dS:-3.32 (3SF)	4	EFE	21754
24	75	0	Inf. Dilution	dS:-0.93 (2SF)	4	EFE	21754
25	100	0	Inf. Dilution	dS:1.6 (2SF)	4	EFE	21754

Reaction No. 72344



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-11.0 (3SF)	0	MSL	140
2	25	0	Inf. Dilution	lgK:-11.82 (4SF)	5	EOD	403
3	25	0	Inf. Dilution	lgK:-11.82 (0.1SD)	3	CRV	6478
4	25	0	Inf. Dilution	lgK:-11.0 (3SF)	0	EFE	13986
5	25	0	Inf. Dilution	lgK:-11.8 (3SF)	4	MSL	19879
6	25	0	Inf. Dilution	lgK:-11.82 (4SF)	4	MSP	21754
7	25	0	Inf. Dilution	lgK:-11.93 (4SF)	5	CRV	31301
8	25	0	Inf. Dilution	lgK:-11.93 (4SF)	5	CRV	31495
9	50	0	Inf. Dilution	lgK:-11.52 (4SF)	4	MSP	21754
10	100	0	Inf. Dilution	lgK:-11.05 (4SF)	4	MSP	21754
11	25	0	Inf. Dilution	dH:15.9 (0.5SD)	0	CRV	6478
12	25	0	Inf. Dilution	dH:4.11 (3SF)	4	MTD	21754
13	50	0	Inf. Dilution	dH:4.80 (3SF)	4	MTD	21754
14	100	0	Inf. Dilution	dH:6.33 (3SF)	4	MTD	21754
15	25	0	Inf. Dilution	dS:-40.4 (3SF)	4	EFE	21754
16	50	0	Inf. Dilution	dS:-37.8 (3SF)	4	EFE	21754
17	100	0	Inf. Dilution	dS:-33.7 (3SF)	4	EFE	21754

Reaction No. 72349



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-4.0 (2SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:-4.24 (3SF)	0	EOD	403
3	25	0	Inf. Dilution	lgK:-4.24 (0.1SD)	0	CRV	6478
4	25	0	Inf. Dilution	dH:7.7 (0.5SD)	0	CRV	6478

Reaction No. 72966

$\text{Sb+3(2)_S-2(3)_(s)} + 2\text{<OH-1>} = \text{Sb+3_S-2(2)} + \text{Sb+3_OH-1(2)_S-2}$							
Note: Both species discredited in later evaluation of original data							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0	Inf. Dilution	$\lg K: -1.10(3\text{SF})$	0	MSL	14010
Note (1): I not kept constant, no electrolyte added (pH 10-11)							
2	25	0	Inf. Dilution	$\lg K: -1.13(3\text{SF})$	1	EAE	

Reaction No. 73013							
$0.5\text{<Sb+3(2)_S-2(3)_(s)>} + 0.5\text{<S-2>} = \text{Sb+3_S-2(2)}$							
Note: The species S-2 does not exist in aqueous solution See reaction: $\text{H+1} + \text{S-2} = \text{H+1_S-2}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0	Inf. Dilution	$\lg K: 0.45(2\text{SF})$	0	MSL	14010
Note (1): I not kept constant, no electrolyte added (pH 8-9)							
2	25	0	Inf. Dilution	$\lg K: 2.(1\text{SF})$	0	ELC	

Reaction No. 73417							
$\text{Sb+3_OH-1(3)_(s)} + \text{H+1_F-1} = \text{Sb+3_F-1_OH-1(2)} + \text{H2O}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	HNO3	$\lg K: -0.37(2\text{SF})$	0	MPL	13997
2	25	0	Inf. Dilution	$\lg K: 3.6(2\text{SF})$	1	ELC	

Reaction No. 73418							
$\text{Sb+3_F-1(3)} + \text{F-1} = \text{Sb+3_F-1(4)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: 2.7(2\text{SF})$	1	EES	
2	30	2	(H,Na)ClO4	$\lg K: 2.7(0.07\text{SD})$	4	MPL	437

Reaction No. 73419							
$\text{Sb+3_F-1(2)} + \text{F-1} = \text{Sb+3_F-1(3)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: 2.6(2\text{SF})$	1	EES	
2	30	2	(H,Na)ClO4	$\lg K: 2.6(0.07\text{SD})$	4	MPL	437

Reaction No. 73420							
$\text{Sb+3_F-1} + \text{F-1} = \text{Sb+3_F-1(2)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

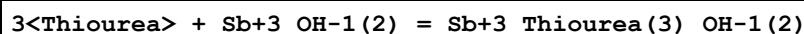
1	25	0	Inf. Dilution	lgK:2.7(2SF)	1	EES	
2	30	2	(H,Na)ClO ₄	lgK:2.7(0.07SD)	4	MPL	437

Reaction No. 73421							
Sb+3_OH-1(2) + 2<H+1> + 3<e-1> = Sb(s) + 2<H2O>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:10.89(4SF)	1	EDH	
2	15	0	Inf. Dilution	E:0.2105(4SF)	4	MEF	5398
3	15	0	Inf. Dilution	E:0.2105(0.0027SD)	5	MEF	13989
4	25	0	Inf. Dilution	E:0.21(2SF)	0	ENS	773
5	25	0	Inf. Dilution	E:0.21(0.01SD)	0	MEF	775
6	25	0	Inf. Dilution	E:0.2040(4SF)	4	MEF	5398
7	25	0	Inf. Dilution	E:0.212(3SF)	5	CRV	6330
8	25	0	Inf. Dilution	E:0.2040(0.0026SD)	5	MEF	13989
9	25	0.2:1.1	HClO ₄	E:0.212(3SF)	5	MEF	7829
10	35	0	Inf. Dilution	E:0.2003(4SF)	4	MEF	5398
11	35	0	Inf. Dilution	E:0.2003(0.0027SD)	2	MEF	13989
12	50	0	Inf. Dilution	E:0.1988(4SF)	2	MEF	5398
13	50	0	Inf. Dilution	dH:0.1988(0.0021SD)	2	MEF	13989

Reaction No. 73422							
2<Thiourea> + Sb+3_OH-1(2) = Sb+3_Thiourea(2)_OH-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	HClO ₄	lgK:0.9(0.1SD)	0	MPS	4114
Note (1): SbO ⁺ in original publication and in all pub. for this reaction							
2	25	0	HClO ₄	lgK:1.(0.1SD)	0	MPS	4125
3	25	0.5	HClO ₄	lgK:2.24(0.02SD)	5	MSL	4111
4	25	1	HClO ₄	lgK:2.36(0.02SD)	5	MSL	4111
5	25	1.5	HClO ₄	lgK:2.5(0.04SD)	2	MSL	4111
6	25	1.5	HClO ₄	lgK:1.8(0.05SD)	0	MPS	4125
7	25	1.8	HClO ₄	lgK:2.2(0.05SD)	2	MPS	4125
8	25	2	HClO ₄	lgK:2.6(0.03SD)	5	MSL	4111
9	25	3	HClO ₄	lgK:2.9(0.04SD)	5	MSL	4111
10	25	3	HClO ₄	lgK:2.8(0.04SD)	5	MCL	4114
11	25	3	HClO ₄	lgK:2.7(0.05SD)	2	MPS	4125
12	25	4	HClO ₄	lgK:3.3(0.06SD)	5	MSL	4111

13	25	4	HClO4	lgK:3.48(0.02SD)	2	MCL	4114
14	25	4	HClO4	lgK:3.9(0.05SD)	0	MPS	4125
15	25	5	HClO4	lgK:4.4(0.03SD)	0	MCL	4114
16	25	0	HClO4	dH:-18.99(4SF)	2	MCL	4114
17	25	3	HClO4	dH:-23.20(4SF)	4	MCL	4114
18	25	4	HClO4	dH:-24.72(4SF)	4	MCL	4114
19	25	5	HClO4	dH:-25.90(4SF)	2	MCL	4114

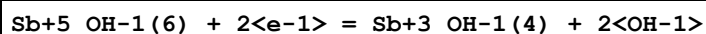
Reaction No. 73423



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	3	H2SO4	lgK:2.29(3SF)	4	MPS	4110

Note (1): SbO+ in original publication

Reaction No. 73424



Note: This reaction is problematic. There are issues with the principal literature source, which we currently assume are typographical (relating to the sign only being explicit at top of each Table). There are also issues with the concentrated KOH medium - various compilations have taken lgK0 from lgK without compunction. lgK0 = -15. by collective decision (MF & PM) 22/7/20. Was -11.8 [EDH] lgK0 = -16 by PMM to match Roper et al. re set of MSb2O6(s) 23/7/20

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-16.(2SF)	1	EES	
2	20	3	KOH	E:-0.428(3SF)	2	MEF	13988

Note (2): Sign assumed negative for all [13988] values from I = 3 to I = 9

3	20	4	KOH	E:-0.484(3SF)	2	MEF	13988
4	20	5	KOH	E:-0.516(3SF)	2	MEF	13988
5	20	6	KOH	E:-0.539(3SF)	2	MEF	13988
6	20	7	KOH	E:-0.554(3SF)	2	MEF	13988
7	20	7.5	KOH	E:-0.561(3SF)	2	MEF	13988
8	20	8	KOH	E:-0.568(3SF)	2	MEF	13988
9	20	9	KOH	E:-0.578(3SF)	2	MEF	13988
10	20	10	KOH	E:-0.6(0.05SD)	2	MEF	775
11	20	10	KOH	E:-0.589(3SF)	0	MEF	13988
12	25	0	Inf. Dilution	E:-0.59(2SF)	0	CRV	6330

Note (12): Appears to have copied value from 25C and high I

13	25	5	KOH	E:-0.516(3SF)	2	EES	
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14	25	10	KOH	E:-0.59 (2SF)	2	ENS	773
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Reaction No. 73425							
$4\text{Sb(s)} + 12\text{OH-1} = \text{Sb}_4\text{O}_6\text{(s)} + 6\text{H}_2\text{O} + 12\text{e-1}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:137.5 (4SF)	1	ELC	
2	25	0	Inf. Dilution	E:0.6831 (0.0013SD)	0	MEF	13990
Note (2): lgK = 138.6; sign changed - see Sb+3_OH-1(4) + 3e-1 = Sb(s) + 4OH-1							

Reaction No. 73426							
$\text{Sb+3_OH-1(2)} + \text{F-1} = \text{Sb+3_F-1_OH-1(2)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	HNO3	lgK:5.5 (2SF)	2	MSL	13997
2	25	0	Inf. Dilution	lgK:5.5 (2SF)	1	ELC	

Reaction No. 73427							
$\text{SbOOH(s)} + \text{H}_2\text{O} = \text{Sb+3_OH-1(2)} + \text{OH-1}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-17.0 (3SF)	0	ENS	5648
Note (1): Secondary reference; unknown solid							

Reaction No. 73428							
$\text{Sb+3_OH-1(2)} + \text{EDTA-4} + 2\text{H+1} = \text{Sb+3_EDTA-4} + 2\text{H}_2\text{O}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:26.1 (3SF)	1	ELC	
2	25	1	NaClO4	lgK:24.8 (3SF)	0	MPS	931
3	25	1	NaClO4	lgK:24.80 (4SF)	0	MSP	14056
4	25	1	Unknown	lgK:22.4 (3SF)	1	EES	

Reaction No. 73429							
$0.5\text{Sb}_2\text{O}_3\text{(rhombic,s)} + 1.5\text{H}_2\text{O} = \text{Sb+3_OH-1(2)} + \text{OH-1}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-17.1 (3SF)	1	EES	
2	25	5	Unknown	lgK:-17.66 (4SF)	0	ENS	773

Reaction No. 73430							
$0.5\text{Sb}_2\text{O}_3\text{(cubic,s)} + 1.5\text{H}_2\text{O} = \text{Sb+3_OH-1(2)} + \text{OH-1}$							

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-17.18(4SF)	1	ELC	
2	25	5	Unknown	lgK:-17.78(4SF)	0	ENS	773

Reaction No. 73431							
$\text{H}_2(\text{g}) + \text{O}_2(\text{g}) + \text{Sb}(\text{s}) - \text{e}^- = \text{Sb}+3_ \text{OH}-1(2)$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:72.2(3SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-99.01(4SF)	2	CRV	7382
Note (2): Modified -42.33 kcal for an additional H ₂ O by adding dGf(H ₂ O) = -56.69 kcal							
3	25	0	Inf. Dilution	dG:-415.2(4SF)kJ	3	EOD	29879

Reaction No. 73432							
$2<\text{H}_2(\text{g})> + 2<\text{O}_2(\text{g})> + \text{Sb}(\text{s}) + \text{e}^- = \text{Sb}+3_ \text{OH}-1(4)$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:142.2(4SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-194.60(5SF)	0	CRV	7382
Note (2): Modified -81.32 kcal by adding 2<H ₂ O> with dGf(H ₂ O) = -56.69 kcal							
3	25	0	Inf. Dilution	dG:-195.78(5SF)	0	CRV	9029
Note (3): Modified -82.40 kcal by adding 2<H ₂ O> with dGf(H ₂ O) = -56.69 kcal							
4	25	0	Inf. Dilution	dH:-246.64(5SF)	0	CRV	9029
Note (4): Modified -110.00 kcal by adding 2<H ₂ O> with dHf(H ₂ O) = -68.32 kcal							

Reaction No. 73433							
$\text{Sb}+3_ \text{Cl}-1(4) + 2<\text{Cl}-1> = \text{Sb}+3_ \text{Cl}-1(6)$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	15	6.3	LiNO ₃	lgK:-0.8(1SF)	0	MDS	931
2	15	6.3	LiNO ₃	lgK:-0.8(1SF)	0	MDS	14051
Note (2): Values given : -1.1 to -0.8							
3	25	0	Inf. Dilution	lgK:-4.5(2SF)	1	ELC	
4	25	4	H ₂ SO ₄	lgK:-0.77(2SF)	0	MSL	14000
5	30	3:12	HCl	lgK:-0.74(2SF)	0	MSP	14001

Reaction No. 73434							
$\text{Sb}+3_ \text{OH}-1(4) + 3<\text{e}^-> = \text{Sb}(\text{s}) + 4<\text{OH}-1>$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-32.0(3SF)	1	ELC	

2	20	10	KOH	E:-0.675(3SF)	0	MEF	13988
Note (2): Measurement in 1923 without sufficient kinetic concern							
3	25	0	Inf. Dilution	E:-0.660(3SF)	0	EFE	600
Note (3): Agrees with Grube and Schweigardt, 1923							
4	25	0	Inf. Dilution	E:-0.6389(4SF)	0	MEF	5398
Note (4): Low wgt to agree with low wgt of original data							
5	25	0	Inf. Dilution	E:-0.66(2SF)	0	CRV	6330
Note (5): No additional information content							
6	25	0	Inf. Dilution	E:-0.6389(0.0013SD)	0	MEF	13990
Note (6): Ref. sign changed							

Reaction No. 73435							
$\text{Sb}_2\text{O}_5(\text{s}) + 6\text{H}^+ + 4\text{e}^- = 2\text{Sb}^{3+} + \text{H}_2\text{O}$							
Note: Sb ₂ O ₅ (s) very probably not present in original experiment							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:40.7(3SF)	1	ELC	
2	25	0	Inf. Dilution	E:0.581(3SF)	0	EFE	600
3	25	0	Inf. Dilution	E:0.581(3SF)	0	CRV	6330

Reaction No. 73436							
$0.5\text{Sb}_2\text{O}_3(\text{s}) + \text{H}^+ + 0.5\text{H}_2\text{O} = \text{Sb}^{3+}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-3.1(2SF)	1	MPL	14050
Note (1): Data from CA 64 13453g							

Reaction No. 73437							
$\text{Sb}^{3+} + 3\text{OH}^- = \text{Sb}(\text{OH})_3(\text{s})$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	12	0.01:0.02	HCl	lgK:-41.5(3SF)	3	MPL	14003
2	12	0	Inf. Dilution	lgK:-41.5(3SF)	3	MPL	140
3	18	0.1	H ₂ SO ₄	lgK:-41.4(3SF)	3	MPL	14012
Note (3): Temperature guess (MF)							
4	25	0.5	KCl	lgK:-43.6(3SF)	0	MPL	14014
Note (4): Same value in Pavlov D Lazarov D Russ. J. Inorg. Chem. 1958, 9, 2099							
5	25	1	(H,Na)ClO ₄	lgK:-41.57(4SF)	5	MSP	13993

Reaction No. 73438							
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0.5<Sb2O3(rhombic,s)> + 1.5<H2O> = Sb+3_OH-1(3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-4.2(2SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:-4.70(3SF)	3	MSL	140
3	25	0	Inf. Dilution	lgK:-4.24(3SF)	4	MSL	8047
4	80	0	Self Medium	lgK:-3.69(0.06SD)	4	MSL	18072

Reaction No. 73439							
0.5<Sb2O3(rhombic,s)> + OH-1 + 1.5<H2O> = Sb+3_OH-1(4)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0:0.1	NaOH	lgK:-2.06(3SF)	0	MSL	8047
2	25	0	Inf. Dilution	lgK:-2.2(2SF)	1	ELC	
3	25	0	Inf. Dilution	lgK:-2.06(3SF)	0	MSL	140

Reaction No. 73440							
Sb+3_OH-1(2) = Sb+3 + 2<OH-1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-29.6(3SF)	1	EDH	
2	25	0.5	KCl	lgK:-47.0(3SF)	0	MPL	14014
Note (2): Same value in Pavlov D Lazarov D Russ. J. Inorg. Chem. 1958, 9, 2099							
3	25	1	(H,Na)ClO4	lgK:-28.62(4SF)	0	MSP	13993

Reaction No. 73441							
Sb+3_OH-1 = Sb+3 + OH-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-15.6(3SF)	1	EDH	
2	25	0.5	KCl	lgK:-46.(2SF)	0	MPL	14014
Note (2): Same value in Pavlov D Lazarov D Russ. J. Inorg. Chem. 1958, 9, 2099							
3	25	1	(H,Na)ClO4	lgK:-14.61(4SF)	0	MSP	13993

Reaction No. 73442							
Sb+3 + 3<OH-1> = Sb+3_OH-1(3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:42.6(3SF)	1	EDH	
2	25	0.5	KCl	lgK:-43.6(3SF)	0	MPL	14014
3	25	1	NaClO4	lgK:41.57(4SF)	0	MSP	13993

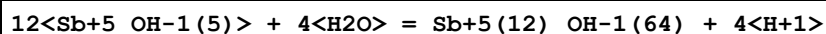
Reaction No. 73443							
Sb+3 + H2O = Sb+3_OH-1 + H+1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.6 (2SF)	1	EDH	
2	25	1	(H,Na)ClO4	lgK:0.61 (2SF)	0	MSP	13993

Reaction No. 73444							
Sb+3_OH-1 + H2O = Sb+3_OH-1(2) + H+1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:0.9 (1SF)	1	EES	
2	25	1	(H,Na)ClO4	lgK:0.013 (2SF)	0	MSP	13993
3	25	5	(H,Na)ClO4	lgK:0.86 (2SF)	6	MSL	7780

Reaction No. 73445							
Sb+5_OH-1(5) + H2O = Sb+5_OH-1(6) + H+1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	10	0	KClO4	lgK:-2.889 (4SF)	5	MPT	23090
2	15	0	KClO4	lgK:-2.871 (4SF)	5	MPT	23090
3	20	0	KClO4	lgK:-2.857 (4SF)	5	MPT	23090
4	25	0	Inf. Dilution	lgK:-2.72 (3SF)	4	EOD	403
5	25	0	Inf. Dilution	lgK:-2.72 (3SF)	3	ENS	773
6	25	0	Inf. Dilution	lgK:-2.72 (3SF)	4	CRV	31301
7	25	0	Inf. Dilution	lgK:-2.72 (3SF)	4	CRV	31495
8	25	0	KClO4	lgK:-2.848 (4SF)	5	MPT	23090
9	25	0	Unknown	lgK:-2.5 (2SF)	1	MPT	25244
10	25	0.5	Me4NCl	lgK:-2.55 (3SF)	4	MPH	14011
11	30	0	KClO4	lgK:-2.841 (4SF)	5	MPT	23090
12	35	0	KClO4	lgK:-2.838 (4SF)	5	MPT	23090
13	40	0	KClO4	lgK:-2.838 (4SF)	5	MPT	23090
14	10	0	KClO4	dH:1.462 (3SF)	5	MTD	23090
15	15	0	KClO4	dH:1.572 (3SF)	5	MTD	23090
16	20	0	KClO4	dH:0.8785 (3SF)	5	MTD	23090
17	25	0	KClO4	dH:0.6176 (3SF)	5	MTD	23090
18	30	0	KClO4	dH:0.3707 (3SF)	5	MTD	23090
19	35	0	KClO4	dH:0.136 (3SF)	5	MTD	23090
20	40	0	KClO4	dH:-0.0910 (3SF)	5	MTD	23090

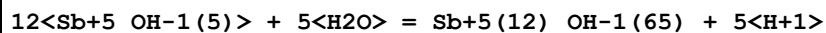
21	10	0	KClO ₄	dS:-7.9646(4SF)	5	EFE	23090
22	15	0	KClO ₄	dS:-9.0339(4SF)	5	EFE	23090
23	20	0	KClO ₄	dS:-9.9933(4SF)	5	EFE	23090
24	25	0	KClO ₄	dS:-10.873(4SF)	5	EFE	23090
25	30	0	KClO ₄	dS:-11.692(4SF)	5	EFE	23090
26	35	0	KClO ₄	dS:-12.462(4SF)	5	EFE	23090
27	40	0	KClO ₄	dS:-13.191(4SF)	5	EFE	23090

Reaction No. 73446



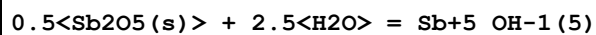
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:20.34(4SF)	6	EOD	403
2	25	0	Inf. Dilution	lgK:20.34(4SF)	5	CRV	31301
3	25	0	Inf. Dilution	lgK:20.34(4SF)	5	CRV	31495

Reaction No. 73447



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:16.72(4SF)	6	EOD	403
2	25	0	Inf. Dilution	lgK:16.72(4SF)	5	CRV	31301
3	25	0	Inf. Dilution	lgK:16.72(4SF)	5	CRV	31495

Reaction No. 73448

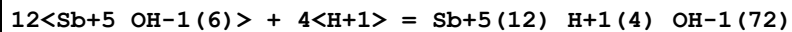


Note: Sb₂O₅(s) probably not the species present but an hydrated gel

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-6.0(2SF)	1	ELC	
2	35	0	Inf. Dilution	lgK:-3.7(2SF)	0	EOD	403

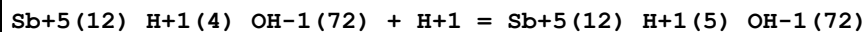
Note (2): Estimated from solubility data from 1948TOU_MOU

Reaction No. 73449



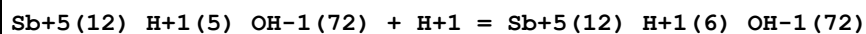
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.5	Me ₄ NC1	lgK:-33.3(3SF)	5	MPH	14011

Reaction No. 73450



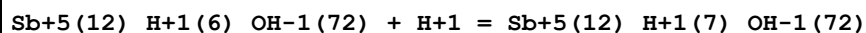
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.5	Me4NC1	lgK:-7.15 (3SF)	5	MPH	14011

Reaction No. 73451



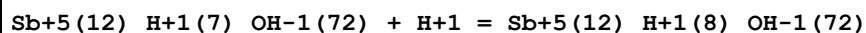
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.5	Me4NC1	lgK:-5.75 (3SF)	5	MPH	14011

Reaction No. 73452



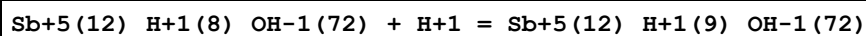
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.5	Me4NC1	lgK:-4.35 (3SF)	5	MPH	14011

Reaction No. 73453



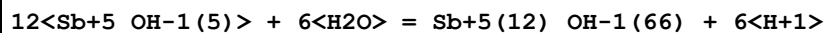
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.5	Me4NC1	lgK:-2.95 (3SF)	5	MPH	14011

Reaction No. 73454



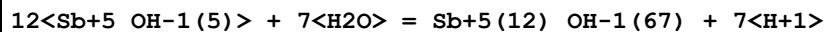
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.5	Me4NC1	lgK:-1.55 (3SF)	5	MPH	14011

Reaction No. 73455



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:11.89 (4SF)	6	EOD	403
2	25	0	Inf. Dilution	lgK:11.89 (4SF)	5	CRV	31301
3	25	0	Inf. Dilution	lgK:11.89 (4SF)	5	CRV	31495

Reaction No. 73456



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:6.07 (3SF)	6	EOD	403
2	25	0	Inf. Dilution	lgK:6.07 (3SF)	5	CRV	31301
3	25	0	Inf. Dilution	lgK:6.07 (3SF)	5	CRV	31495

Reaction No. 73457

Sb+5_OH-1(6) + 2<H+1> = Sb+5_OH-1(4) + 2<H2O>							
Note: Does Sb+5_OH-1(4) exist?							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-0.54 (2SF)	0	EFE	13986

Reaction No. 73458							
Sb+5_Cl-1(4)_OH-1(2) + H+1 + Cl-1 = Sb+5_Cl-1(5)_OH-1 + H2O							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	26	6:9	HCl	lgK:-3.43 (3SF)	0	MSP	13991

Reaction No. 73459							
Sb+5_Cl-1(5)_OH-1 + H+1 + Cl-1 = Sb+5_Cl-1(6) + H2O							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	26	8:10	HCl	lgK:-4.65 (3SF)	0	MSP	13991

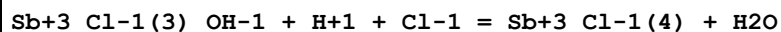
Reaction No. 73460							
Sb(s) + 3<S-2> = Sb+3_S-2(3) + 3<e-1>							
Note: The species S-2 does not exist in aqueous solution See reaction: H+1 + S-2 = H+1_S-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:36.8 (3SF)	0	ELC	
2	25	0	Inf. Dilution	E:0.830 (3SF)	0	MEF	14048
Note (2): Sign changed (otherwise even worse); inexplicable agreement with Latimer							

Reaction No. 73461							
Sb+5 + 2<e-1> = Sb+3							
Note: Sb+5 is now considered not to exist (even in strong acid)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	3.5	HCl	E:-0.75 (2SF)	0	ENS	773
2	25	3.5	HCl	E:-0.746 (3SF)	0	MEF	13987
3	25	4.5	HCl	E:-0.784 (3SF)	0	MEF	13987
4	25	6	HCl	E:-0.818 (3SF)	0	MEF	13987

Reaction No. 73462							
SbOCl(s) + H+1 + 2<Cl-1> = Sb+3_Cl-1(3)_OH-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	6	(H,Na)ClO4	lgK:-1.2 (2SF)	4	MSL	14050

Note (1): Data taken from CA 64 13453g

Reaction No. 73463



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	6	(H,Na)ClO4	lgK:0.18(2SF)	4	MSL	14050

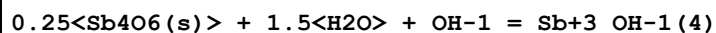
Note (1): Data taken from CA 64 13453g

Reaction No. 73464



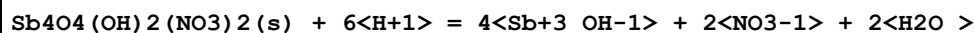
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	15	0	Inf. Dilution	E:0.1545(4SF)	3	MEF	5398
2	15	0	Inf. Dilution	E:0.1545(0.0018SD)	3	MEF	13989
3	25	0	Inf. Dilution	E:0.1504(4SF)	5	MEF	5398
4	25	0	Inf. Dilution	E:0.1504(0.0005SD)	5	MEF	13989
5	35	0	Inf. Dilution	E:0.1468(4SF)	4	MEF	5398
6	35	0	Inf. Dilution	E:0.1468(0.0016SD)	5	MEF	13989
7	50	0	Inf. Dilution	E:0.1391(4SF)	3	MEF	5398
8	50	0	Inf. Dilution	E:0.1391(0.0028SD)	3	MEF	13989

Reaction No. 73465



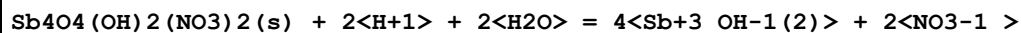
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	15	0	Inf. Dilution	lgK:-2.44(3SF)	3	MSL	5398
2	25	0	Inf. Dilution	lgK:-2.35(3SF)	3	MSL	5398
3	35	0	Inf. Dilution	lgK:-2.21(3SF)	3	MSL	5398
4	50	0	Inf. Dilution	lgK:-2.08(3SF)	3	MSL	5398

Reaction No. 73466



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-17.7(3SF)	1	EES	
2	25	5	(H,Na)NO3	lgR:-16.89(4SF)	5	MSL	7780
3	25	5	NaNO3	lgR:-16.89(4SF)	3	MSL	5398

Reaction No. 73467



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-14.5(3SF)	1	EES	
2	25	5	(H,Na)NO3	lgK:-13.46(4SF)	5	MSL	7780
3	25	5	NaNO3	lgR:-13.46(4SF)	3	MSL	5398

Reaction No. 73468							
Sb4O5(OH)ClO4(H2O)0.5(s) + 3<H+1> + 1.5<H2O> = 4<Sb+3_OH-1(2)> + ClO4-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	5	(H,Na)ClO4	lgK:-13.07(4SF)	5	MSL	7780
2	25	5	NaClO4	lgR:-13.07(4SF)	3	MSL	5398

Reaction No. 73469							
Sb+3(2)_S-2(3)_ (s) + 2<H2O> = Sb+3(2)_OH-1(2)_S-2(2) + H+1(2)_S-2							
Note: This species does not form at 25C							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-40.(2SF)	1	EAE	
Note (1): Guess to deal with extrapolation problem							
2	200	0.1	Mixed salts	lgK:-7.44(0.25SD)	0	MSL	14007
Note (2): I not constant, I values given in paper							
3	275	0.1	Mixed salts	lgK:-5.92(0.15SD)	0	MSL	14007
Note (3): I not constant, I values given in paper							
4	350	0.1	Mixed salts	lgK:-4.73(0.09SD)	0	MSL	14007
Note (4): I not constant, I values given in paper							

Reaction No. 73470							
Sb+3(2)_H+1(2)_S-2(4) = Sb+3(2)_H+1_S-2(4) + H+1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-4.9(2SF)	1	EES	
2	25	0	Inf. Dilution	lgK:-2.31(3SF)	2	EFE	14060
3	25	0	Inf. Dilution	lgK:-2.4(2SF)	2	EOD	14061
Note (3): I not given, probably not constant							
4	25	0.1	Mixed salts	lgK:-4.91(3SF)	2	MSL	14007
Note (4): I not constant, I values given in paper							
5	30	0	Inf. Dilution	lgK:-2.09(3SF)	1	EFE	14060
6	50	0	Inf. Dilution	lgK:-1.30(3SF)	1	EFE	14060
7	70	0	Inf. Dilution	lgK:-0.60(2SF)	1	EFE	14060
8	90	0	Inf. Dilution	lgK:0.02(1SF)	0	EFE	14060

9	90	0.1	Mixed salts	lgK:-3.56(3SF)	0	MSL	14007
Note (9): I not constant, I values given in paper							
10	100	0	Inf. Dilution	lgK:0.30(2SF)	1	EFE	14060
11	200	0.1	Mixed salts	lgK:-5.64(3SF)	1	MSL	14007
Note (11): I not constant, I values given in paper							

Reaction No. 73471							
Sb+3(2)_H+1_S-2(4) = Sb+3(2)_S-2(4) + H+1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-9.5(2SF)	1	EES	
2	25	0	Inf. Dilution	lgK:-11.50(4SF)	2	EOD	14006
3	25	0.1	Mixed salts	lgK:-9.52(3SF)	2	MSL	14007
Note (3): I not constant, I values given in paper							
4	50	0	Inf. Dilution	lgK:-11.30(4SF)	0	EOD	14006
5	90	0.1	Mixed salts	lgK:-8.42(3SF)	0	MSL	14007
Note (5): I not constant, I values given in paper							
6	100	0	Inf. Dilution	lgK:-11.00(4SF)	0	EOD	14006
7	150	0	Inf. Dilution	lgK:-10.80(4SF)	0	EOD	14006
8	200	0	Inf. Dilution	lgK:-10.65(4SF)	0	EOD	14006
9	200	0.1	Mixed salts	lgK:-7.57(3SF)	0	MSL	14007
Note (9): I not constant, I values given in paper							
10	250	0	Inf. Dilution	lgK:-10.50(4SF)	0	EOD	14006
11	300	0	Inf. Dilution	lgK:-10.40(4SF)	0	EOD	14006
12	25	0.1	Mixed salts	dH:29.892(5SF)kJ	0	ETR	14007
Note (12): I not constant, I values given in paper							
13	90	0.1	Mixed salts	dH:28.761(5SF)kJ	0	ETR	14007
Note (13): I not constant, I values given in paper							
14	200	0.1	Mixed salts	dH:29.773(5SF)kJ	0	ETR	14007
Note (14): I not constant, I values given in paper							

Reaction No. 73472							
Sb+3(2)_S-2(3)_(s) + H+1_S-2 + H+1 = Sb+3(2)_H+1(2)_S-2(4)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.8(2SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:0.25(0.7SD)	0	EOD	14006

Reaction No. 73473							
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Sb+3(2)_S-2(3)_(s) + H+1_S-2 = Sb+3(2)_S-2(4) + H+1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-12.12(4SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:-14.00(0.2SD)	0	EOD	14006
3	30	0	Self Medium	lgK:-13.33(0.03SD)	0	MSL	29949
4	50	0	Inf. Dilution	lgK:-13.10(0.2SD)	0	EOD	14006
5	100	0	Inf. Dilution	lgK:-11.80(0.2SD)	0	EOD	14006
6	150	0	Inf. Dilution	lgK:-11.20(0.2SD)	0	EOD	14006
7	200	0	Inf. Dilution	lgK:-10.90(0.2SD)	0	EOD	14006
8	250	0	Inf. Dilution	lgK:-10.65(0.2SD)	0	EOD	14006
9	300	0	Inf. Dilution	lgK:-10.50(0.2SD)	0	EOD	14006

Reaction No. 73474							
Sb+3(2)_S-2(3)_(s) + 6<H2O> = 2<Sb+3_OH-1(3)> + 3<H+1(2)_S-2>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-40.5(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:-37.31(4SF)	0	EFE	14060
3	30	0	Inf. Dilution	lgK:-36.54(4SF)	0	EFE	14060
4	50	0	Inf. Dilution	lgK:-33.69(4SF)	0	EFE	14060
5	70	0	Inf. Dilution	lgK:-31.18(4SF)	0	EFE	14060
6	90	0	Inf. Dilution	lgK:-28.94(4SF)	0	EFE	14060
7	100	0	Inf. Dilution	lgK:-27.91(4SF)	0	EFE	14060

Reaction No. 73595							
Sb+3_OH-1(2) + HOEDTA-3 + H+1 = Sb+3_OH-1_HOEDTA-3 + H2O							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:18.7(3SF)	1	ELC	
2	25	1	NaClO4	lgK:20.18(4SF)	0	MGL	14055

Reaction No. 74533							
Gly-1 + Sb+3 = Sb+3_Gly-1							
Note: Sb+3 is not what the authors had in solution; does it exist?							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	30	0.1	KCl	lgK:10.6(3SF)	0	MPL	22636

Reaction No. 74534							
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Sb+3 + 2<Maltol-1> = Sb+3_Maltol-1(2)							
Note: Sb+3 is not what the authors had in solution; does it exist?							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	30	0.1	KCl	lgK:21.3(3SF)	0	MPL	22637

Reaction No. 74535							
Sb+3 + 2<Kojic-1> = Sb+3_Kojic-1(2)							
Note: Sb+3 is not what the authors had in solution; does it exist?							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	30	0.1	KCl	lgK:19.7(3SF)	0	MPL	22637

Reaction No. 75185							
2<Sb+3_OH-1(4)> + 2<H+1> = Sb2O3(s) + 5<H2O>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:31.6(3SF)	1	ELC	
2	25	0	H2SO4	lgK:32.(2SF)	0	MSL	23088
Note (2): EBP: 0.5<Sb2O3(s)> + 3<H+1> + 3<e-1> = Sb(s) + 1.5<H2O>							

Reaction No. 75186							
Sb+3 + 2<ClKojic-1> = Sb+3_ClKojic-1(2)							
Note: Sb+3 is not what the authors had in solution; does it exist?							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	30	0.1	KCl	lgK:18.3(3SF)	0	MPL	22637

Reaction No. 75187							
Sb+3_H+1(-1)_OH-1_Citric-3 + OH-1 + H2O = Sb+3_OH-1(3) + Citric-3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	0	Unknown	lgK:-8.76(3SF)	0	MSL	22956
Note (1): I not kept constant, no electrolyte added. Doubtful value; serious doubts in [23085]							
2	25	0	Inf. Dilution	lgK:2.9(2SF)	1	ELC	

Reaction No. 75188							
Sb+3_OH-1(3) + Lactic-1 = Sb+3_H+1(-1)_Lactic-1_OH-1(2) + H2O							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0	Unknown	lgK:1.89(3SF)	5	MPH	23085
2	25	0	Inf. Dilution	lgK:1.89(3SF)	1	ESC	

Reaction No. 75189							
$\text{Sb+3_H+1(-1)_Citric-3} + \text{H2O} = \text{Sb+3_H+1(-1)_OH-1_Citric-3} + \text{H+1}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	0	Unknown	lgK:-4.45(3SF)	0	MSL	22956
Note (1): I not kept constant, no electrolyte added; serious doubts in [23085]							
2	25	0	Inf. Dilution	lgK:-4.45(3SF)	1	EAE	

Reaction No. 75190							
$\text{Sb+5_OH-1(6)} + \text{H+1_Citric-3} + \text{H+1} = \text{Sb+5_OH-1(4)_Citric-3} + 2\text{<H2O>}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	KNO3	lgK:7.8(0.2SD)	4	MGL	25244

Reaction No. 75191							
$\text{Sb+3_OH-1(3)} + \text{H+1(2)_Citric-3} = \text{Sb+3_OH-1(2)_Citric-3} + \text{H2O} + \text{H+1}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0	Unknown	lgK:0.1(1SF)	5	MPH	23085
2	25	0	Inf. Dilution	lgK:-0.0187(3SF)	2	EAE	
3	60	0	Unknown	lgK:-1.1(2SF)	4	MPH	23085
4	20:60	0	Unknown	dH:-15.(2SF)	3	MTD	23085
5	20:60	0	Unknown	dS:-48.(2SF)	3	EFE	23085

Reaction No. 75192							
$\text{Sb+3_OH-1(2)} + \text{H+1_Tartaric-2} = \text{Sb+3_OH-1(2)_Tartaric-2} + \text{H+1}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	0	Unknown	lgK:4.39(3SF)	0	MSL	23005
Note (1): I not ct, noelectr. added; unknown tartaric type; Sb(OH)2 not main species at working pH							
2	25	0	Inf. Dilution	lgK:4.3(2SF)	1	EES	

Reaction No. 75193							
$\text{Sb+3_OH-1(2)} + \text{H+1(2)_Citric-3} = \text{Sb+3_Citric-3} + 2\text{<H2O>}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	0	Unknown	lgK:5.22(3SF)	0	MSL	22956
Note (1): I not ct; no electr added; serious doubts in [23085]; Sb(OH)2 not main species at working pH							
2	25	0	Inf. Dilution	lgK:5.22(3SF)	1	EAE	

Reaction No. 75195							
$2\text{Sb}+5\text{OH}^{-1}(6) + \text{Pb}^{+2} = \text{Pb}[\text{Sb}(\text{OH})_6]_2(\text{s})$ Note: Pb antimonates incongruent dissolution; see comment in equivalent reaction with $\text{Ca}[\text{Sb}(\text{OH})_6]_2(\text{s})$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Unknown	lgK:11.02 (4SF)	2	MSL	23091

Reaction No. 75196							
$2\text{Sb}+5\text{OH}^{-1}(6) + \text{Mg}^{+2} + 6\text{H}_2\text{O} = \text{Mg}(\text{H}_2\text{O})_6. [\text{Sb}(\text{OH})_6]_2(\text{s})$ Note: Mineral name: brandholzite							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:7.74 (3SF)	5	MSL	23089
Note (1): Authors also gave delta G of formation							

Reaction No. 75197							
$\text{Sb}+3\text{OH}^{-1}(2) + 2\text{H}^{+1}\text{Oxalic-2} = \text{Sb}+3\text{Oxalic-2}(2) + 2\text{H}_2\text{O}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:6.8 (2SF)	0	ESC	
Note (1): I wg = 0 because lgK insists in taking this value instead of Tella							
2	25	0	Unknown	lgK:6.8 (2SF)	0	MSL	22957
Note (2): I not kept constant, no electrolyte added; Sb(OH) ₂ not main form at working pH							
3	27	0	Unknown	lgK:7.1 (2SF)	0	MSL	22957
Note (3): I not kept constant, no electrolyte added; Sb(OH) ₂ not main form at working pH							
4	30	0	Unknown	lgK:7.5 (2SF)	0	MSL	22957
Note (4): I not kept constant, no electrolyte added; Sb(OH) ₂ not main form at working pH							

Reaction No. 75198							
$\text{Sb}+3\text{OH}^{-1}(3) + 2\text{H}^{+1}(2)\text{Citric-3} = \text{Sb}+3\text{H}^{+1}(2)\text{Citric-3}(2) + 2\text{H}_2\text{O} + \text{OH}^{-1}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0	Unknown	lgK:-3.9 (2SF)	5	MPH	23085
2	25	0	Inf. Dilution	lgK:-4.17 (3SF)	2	EAE	
3	60	0	Unknown	lgK:-5.6 (2SF)	4	MPH	23085
4	20:60	0	Unknown	dH:-18. (2SF)	3	MTD	23085
5	20:60	0	Unknown	dS:-79. (2SF)	3	EFE	23085

Reaction No. 75199							
$2\text{Sb}+5\text{OH}-1(6) + \text{Ni}+2 + 6\text{H}_2\text{O} = \text{Ni}(\text{H}_2\text{O})_6.[\text{Sb}(\text{OH})_6]_2(\text{s})$							
Note: Mineral name: bottinoite							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:9.89(3SF)	5	MSL	23089
Note (1): Authors also give deltaG of formation							

Reaction No. 75200							
$2\text{Sb}+3\text{OH}-1(4) + 3\text{SO}_4-2 + 8\text{H}+1 = \text{Sb}+3(2)\text{SO}_4-2(3)(\text{s}) + 8\text{H}_2\text{O}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:18.5(3SF)	1	ELC	
2	25	0	H ₂ SO ₄	lgK:55.(2SF)	0	MSL	23088

Reaction No. 75201							
$\text{Sb}+5\text{Na}+1\text{OH}-1(6)(\text{s}) = \text{Sb}+5\text{OH}-1(6) + \text{Na}+1$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	18	0	Self Medium	lgK:-5.325(4SF)	2	MSL	31661
2	25	0	Self Medium	lgK:-5.017(4SF)	3	MSL	31657
Note (2): Calculated by Lemire 1999 from data by Blandamer et al 1964							
3	25	0	Self Medium	lgK:-5.072(0.050SD)	3	MSL	31657
Note (3): Solid A							
4	25	0	Self Medium	lgK:-5.256(0.046SD)	3	MSL	31657
Note (4): Solid C							
5	25	0	Self Medium	lgK:-4.551(4SF)	2	MSL	31659
6	25	0	Self Medium	lgK:-5.100(4SF)	2	MSL	31661
7	25	0	Unknown	lgK:-5.05(3SF)	3	MSL	23089
Note (7): Calc. from sol. data in Blandamer et al JCS 1974 1084							
8	33.5	0	Self Medium	lgK:-4.831(4SF)	2	MSL	31661
9	35	0	Self Medium	lgK:-4.776(4SF)	2	MSL	31657
Note (9): Calculated by Lemire 1999 from data by Blandamer et al. 1964							
10	50	0	Self Medium	lgK:-4.518(4SF)	2	MSL	31567
Note (10): Calculated by Lemire 1999 from data by Blandamer et al. 1964							
11	50	0	Self Medium	lgK:-3.648(4SF)	1	MSL	31657
Note (11): Calculated by Lemire 1999 from data by Urazov							
12	70	0	Self Medium	lgK:-4.241(4SF)	2	MSL	31567
Note (12): Calculated by Lemire 1999 from data by Blandamer et al. 1964							

13	75	0	Self Medium	lgK:-3.046(4SF)	2	MSL	31567
Note (13): Calculated by Lemire 1999 from data by Urazov							
14	80	0	Self Medium	lgK:-4.159(4SF)	2	MSL	31567
Note (14): Calculated by Lemire 1999 from data by Blandamer et al. 1964							
15	20:80	0	Unknown	dH:7.9(2SF)	3	MTD	23087

Reaction No. 75202							
Sb+3_Citric-3 = Sb+3_H+1(-1)_Citric-3 + H+1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	0	Unknown	lgK:-2.72(3SF)	0	MSL	22956
Note (1): I not kept constant, no electrolyte added; serious doubts in [23085]							
2	25	0	Inf. Dilution	lgK:-2.1(2SF)	1	ELC	

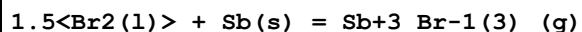
Reaction No. 75203							
Sb+3_OH-1(3) + H+1(2)_Citric-3 = Sb+3_H+1_OH-1(2)_Citric-3 + H2O							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0	Unknown	lgK:4.6(2SF)	5	MPH	23085
2	25	0	Inf. Dilution	lgK:4.6(2SF)	1	ESC	
3	60	0	Unknown	lgK:4.0(2SF)	3	MPH	23085
4	20:60	0	Unknown	dH:-7.7(2SF)	3	MTD	23085
5	20:60	0	Unknown	dS:-2.(1SF)	3	EFE	23085

Reaction No. 75204							
Sb+3_OH-1(3) + 2<H+1_Oxalic-2> = Sb+3_Oxalic-2(2) + OH-1 + 2<H2O>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0	Unknown	lgK:-5.9(2SF)	5	MPH	23085
2	25	0	Inf. Dilution	lgK:-5.9(2SF)	1	ESC	
3	60	0	Unknown	lgK:-6.3(2SF)	4	MPH	23085
4	20:60	0	Unknown	dH:-4.5(2SF)	3	MTD	23085

Reaction No. 75205							
Sb+3_OH-1(3) + H+1_Oxalic-2 = Sb+3_OH-1(2)_Oxalic-2 + H2O							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0	Unknown	lgK:3.8(2SF)	5	MPH	23085
2	25	0	Inf. Dilution	lgK:3.8(2SF)	1	ESC	
3	60	0	Unknown	lgK:2.9(2SF)	4	MPH	23085
4	20:60	0	Unknown	dH:-8.8(2SF)	3	MTD	23085

5	20:60	0	Unknown	dS:-14.(2SF)	3	EFE	23085
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Reaction No. 75592



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-223.9(4SF)kJ	3	CRV	6330
2	25	0	Inf. Dilution	dH:-194.6(4SF)kJ	3	CRV	6330

Reaction No. 75924

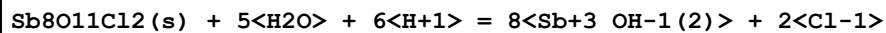


Note: Mineral name: klebelsbergite

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1084	HNO3	lgK:-25.57(0.10SD)	4	MSL	30103

Note (1): Authors also give deltaGf

Reaction No. 75989

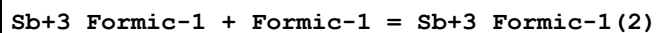


Note: Mineral name: onoratoite

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1055	HCl	lgK:-42.2(0.2SD)	4	MSL	30556

Note (1): Authors also give deltaG of formation

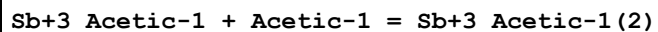
Reaction No. 76246



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.7	NaClO4	lgK:4.93(3SF)	0	MVM	3009

Note (1): Based on presence of Sb+3 in exp. solution, wrong

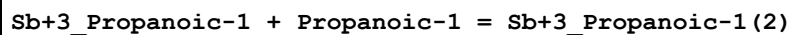
Reaction No. 76266



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.7	NaClO4	lgK:5.64(3SF)	0	MVM	3009

Note (1): Based on presence of Sb+3 in exp. solution, wrong

Reaction No. 76295



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
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1	20	0.7	NaClO4	lgK:4.52 (3SF)	0	MVM	3009
Note (1): Based on presence of Sb+3 in exp. solution, wrong							

Reaction No. 76323							
Sb+3_Glutaric-2 + Glutaric-2 = Sb+3_Glutaric-2 (2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.7	NaClO4	lgK:9.30 (3SF)	0	MVM	3009
Note (1): Based on presence of Sb+3 in exp. solution, wrong							

Reaction No. 76992							
4<S(s)> + 2<Sb(s)> + 2<e-1> = Sb+3(2)_S-2(4)							
Note: q							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:16.4 (3SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-99.5 (3SF) kJ	0	CRV	9015
3	25	0	Inf. Dilution	dG:-99.5 (3SF) kJ	0	EOD	29489
Note (3): Copied from Wagman1982 [9015]							
4	25	0	Inf. Dilution	dG:-23.8 (3SF)	0	EFE	31675
5	25	0	Inf. Dilution	dH:-219.2 (4SF) kJ	0	CRV	9015
6	25	0	Inf. Dilution	dH:-219.2 (4SF) kJ	0	EOD	29489

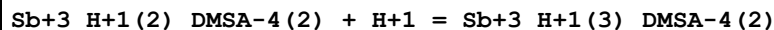
Reaction No. 77311							
Sb+3_OH-1_DMSA-4 + H+1 = Sb+3_DMSA-4 + H2O							
Note: Wrong ligand, DMSA-4 in original reaction (7217) is dlDMSA-4							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	Unknown	lgK:4.90 (3SF)	0	CRV	7271

Reaction No. 77312							
Sb+3_DMSA-4 (2) + H+1 = Sb+3_H+1_DMSA-4 (2)							
Note: Wrong ligand, DMSA-4 in original reaction (11022) is dlDMSA-4							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	Unknown	lgK:6.82 (3SF)	0	CRV	7271

Reaction No. 77313							
Sb+3_H+1_DMSA-4 (2) + H+1 = Sb+3_H+1 (2)_DMSA-4 (2)							
Note: Wrong ligand, DMSA-4 in original reaction (11021) is dlDMSA-4							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

1	20	0.1	Unknown	lgK:4.61(3SF)	0	CRV	7271
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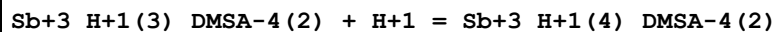
Reaction No. 77314



Note: Wrong ligand, DMSA-4 in original reaction (11020) is d1DMSA-4

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	Unknown	lgK:3.60(3SF)	0	CRV	7271

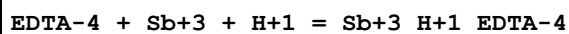
Reaction No. 77315



Note: Wrong ligand, DMSA-4 in original reaction (11019) is d1DMSA-4

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	Unknown	lgK:2.57(3SF)	0	CRV	7271

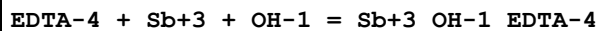
Reaction No. 78275



Note: Not a good idea to estimate reactions with Sb3+, weak link between Sb3+ and species existing in solution

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:28.7(3SF)	0	ELC	

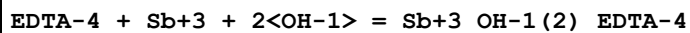
Reaction No. 78501



Note: Not a good idea to estimate reactions with Sb3+, weak link between Sb3+ and species existing in solution

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:35.46(4SF)	0	ELC	

Reaction No. 78523



Note: Not a good idea to estimate reactions with Sb3+, weak link between Sb3+ and species existing in solution

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:41.38(4SF)	0	ELC	

Note (1): MF: wg=0 to avoid the reaction misleading LC. Right approach?

Reaction No. 78927

$3\text{SO}_4\text{-2} + 2\text{Sb+3} = \text{Sb+3(2)}_{\text{SO}_4\text{-2(3)}}(\text{s})$							
Note: Not a good idea to estimate reactions with Sb3+, weak link between Sb3+ and species existing in solution							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-3.2(2SF)	1	ELC	

Reaction No. 79310							
$\text{SO}_4\text{-2} + \text{Sb+3} + 2\text{OH-1} = \text{Sb+3}_{\text{OH-1(2)}}_{\text{SO}_4\text{-2}}$							
Note: Not a good idea to estimate reactions with Sb3+, weak link between Sb3+ and species existing in solution							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:29.98(4SF)	1	ELC	

Reaction No. 79392							
$2\text{Oxalic-2} + \text{Sb+3} = \text{Sb+3}_{\text{Oxalic-2(2)}}$							
Note: Not a good idea to estimate reactions with Sb3+, weak link between Sb3+ and species existing in solution							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:16.5(3SF)	0	ELC	

Reaction No. 79393							
$\text{Citric-3} + \text{Sb+3} + \text{OH-1} - \text{H+1} = \text{Sb+3}_{\text{H+1(-1)}}_{\text{OH-1}}_{\text{Citric-3}}$							
Note: Not a good idea to estimate reactions with Sb3+, weak link between Sb3+ and species existing in solution							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:25.1(3SF)	0	ELC	

Reaction No. 79394							
$\text{Oxalic-2} + \text{Sb+3} + 2\text{OH-1} = \text{Sb+3}_{\text{OH-1(2)}}_{\text{Oxalic-2}}$							
Note: Not a good idea to estimate reactions with Sb3+, weak link between Sb3+ and species existing in solution							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:35.9(3SF)	0	ELC	

Reaction No. 79395							
$\text{Citric-3} + \text{Sb+3} - \text{H+1} = \text{Sb+3}_{\text{H+1(-1)}}_{\text{Citric-3}}$							
Note: Not a good idea to estimate reactions with Sb3+, weak link between Sb3+ and species existing in solution							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

1	25	0	Inf. Dilution	lgK:15.6(3SF)	0	ELC	
Note (1): MF: wg=0 to avoid the reaction misleading LC. Right approach?							

Reaction No. 79396							
2<Citric-3> + Sb+3 + 2<H+1> = Sb+3_H+1(2)_Citric-3(2)							
Note: Not a good idea to estimate reactions with Sb3+, weak link between Sb3+ and species existing in solution							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:32.04(4SF)	0	ELC	

Reaction No. 79397							
Citric-3 + Sb+3 = Sb+3_Citric-3							
Note: Not a good idea to estimate reactions with Sb3+, weak link between Sb3+ and species existing in solution							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:17.65(4SF)	0	ELC	

Reaction No. 79702							
Sb+3_DTPA-5 + 2<OH-1> = Sb+3_OH-1(2)_DTPA-5							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	NaClO4	lgK:9.82(3SF)	2	CRV	13242
2	25	0	Inf. Dilution	lgK:10.0(3SF)	1	EES	

Reaction No. 79703							
DTPA-5 + Sb+3 = Sb+3_DTPA-5							
Note: Not a good idea to estimate reactions with Sb3+, weak link between Sb3+ and species existing in solution							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:28.(2SF)	0	EES	
Note (1): Guess based on Al+3 and Bi+3 and LC							

Reaction No. 79806							
0.5<F2(g)> + H2(g) + O2(g) + Sb(s) = Sb+3_F-1_OH-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-724.6(4SF)kJ	3	EFE	9015
2	25	0	Inf. Dilution	dG:-173.2(4SF)	4	CRV	12011
3	25	0	Inf. Dilution	dG:-726.6(4SF)kJ	2	EOD	29489
Note (3): Copied from Wagman1982 [9015]							

Reaction No. 79807							
$3\text{H}_2(\text{g}) + 3\text{O}_2(\text{g}) + \text{Sb}(\text{s}) + \text{e}^- = \text{Sb}_5\text{OH}^{-1}(\text{g})$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: 209.2(4\text{SF})$	1	ELC	
2	25	0	Inf. Dilution	$\text{dG}: -1188.70(6\text{SF}) \text{ kJ}$	0	EOD	29489

Reaction No. 79808							
$1.5\text{O}_2(\text{g}) + 2\text{Sb}(\text{s}) = \text{Sb}_2\text{O}_3(\text{s})$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: 108.3(4\text{SF})$	1	ELC	
2	25	0	Inf. Dilution	$\text{dG}: -626.35(5\text{SF}) \text{ kJ}$	0	EOD	29489
3	25	0	Inf. Dilution	$\text{dG}: -149.76(5\text{SF})$	0	CRV	31291
4	25	0	Inf. Dilution	$\text{dH}: -708.56(5\text{SF}) \text{ kJ}$	2	EOD	29489

Reaction No. 79809							
$3\text{O}_2(\text{g}) + 4\text{Sb}(\text{s}) = \text{Sb}_4\text{O}_6(\text{cubic}, \text{s})$							
Note: This solid is Sb ₂ O ₃ but Wagman & Co started writing it as Sb ₄ O ₆ ; this has created confusion in databases							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\text{dG}: -1268.1(5\text{SF}) \text{ kJ}$	0	EFE	9015
2	25	0	Inf. Dilution	$\text{dG}: -1268.1(5\text{SF}) \text{ kJ}$	0	EOD	29489
Note (2): Copied from Wagman1982 [9015]							
3	25	0	Inf. Dilution	$\text{dG}: -303.1(4\text{SF})$	0	EFE	31675
4	25	0	Inf. Dilution	$\text{dH}: -1440.6(5\text{SF}) \text{ kJ}$	2	EOD	29489

Reaction No. 79810							
$3\text{O}_2(\text{g}) + 4\text{Sb}(\text{s}) = \text{Sb}_4\text{O}_6(\text{rhombic}, \text{s})$							
Note: This solid is Sb ₂ O ₃ but Wagman & Co started writing it as Sb ₄ O ₆ ; this has created confusion in databases							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: 218.2(4\text{SF})$	1	ELC	
2	25	0	Inf. Dilution	$\text{dG}: -1253.0(5\text{SF}) \text{ kJ}$	0	EFE	9015
3	25	0	Inf. Dilution	$\text{dG}: -1253.0(5\text{SF}) \text{ kJ}$	0	EOD	29489
Note (3): Copied from Wagman1982 [9015]							
4	25	0	Inf. Dilution	$\text{dG}: -299.5(4\text{SF})$	0	EFE	31675
5	25	0	Inf. Dilution	$\text{dH}: -1417.1(5\text{SF}) \text{ kJ}$	2	EOD	29489

Reaction No. 80005							
$2\text{Sb}+5\text{OH}^{-1}(6) + \text{Ca}^{+2} = \text{Ca}[\text{Sb}(\text{OH})_6]_2(\text{s})$							
Note: Ca antimonate (romeite) incongruent dissolution; Ca:Sb ratio in solid depends on Ca present							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Self Medium	$\lg K: 9.8(2\text{SF})$	3	MSL	31676
Note (1): most soluble case							
2	25	0	Self Medium	$\lg K: 9.57(0.08\text{SD})$	3	MSL	31677
Note (2): most soluble case							
3	25	0	Unknown	$\lg K: 12.55(4\text{SF})$	0	MSL	23091
Note (3): possibly a bit aged							

Reaction No. 80006							
$0.5\text{Sb}_2\text{O}_3(\text{cubic}, \text{s}) + 1.5\text{H}_2\text{O} = \text{Sb}+3\text{OH}^{-1}(3)$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	80	0	Self Medium	$\lg K: -3.96(0.08\text{SD})$	2	MSL	18072

Reaction No. 80007							
$0.25\text{Sb}_4\text{O}_6(\text{rhombic}, \text{s}) + 1.5\text{H}_2\text{O} + \text{OH}^{-1} = \text{Sb}+3\text{OH}^{-1}(4)$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	15	0	Inf. Dilution	$\lg K: -2.44(0.02\text{SD})$	4	MSL	14047
2	25	0	Inf. Dilution	$\lg K: -2.35(0.02\text{SD})$	4	MSL	14047
3	25	2	NaClO ₄	$\lg K: -2.22(3\text{SF})$	4	MSL	14047
4	35	2	Inf. Dilution	$\lg K: -2.21(0.02\text{SD})$	4	MSL	14047
5	50	0	Inf. Dilution	$\lg K: -2.08(0.02\text{SD})$	4	MSL	14047

Reaction No. 80008							
$\text{Sb}_2\text{O}_3(\text{rhombic}, \text{s}) + 3\text{H}_2\text{O} = 2\text{Sb}+3\text{OH}^{-1}(3)$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: -8.72(3\text{SF})$	5	CRV	31301
2	25	0	Inf. Dilution	$\lg K: -8.72(3\text{SF})$	5	CRV	31495

Reaction No. 80009							
$\text{Sb}+3\text{OH}^{-1}(3) + 2\text{H}^{+1} = \text{Sb}+3\text{OH}^{-1} + 2\text{H}_2\text{O}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: 0.83(2\text{SF})$	1	CRV	31301
2	25	0	Inf. Dilution	$\lg K: 0.83(2\text{SF})$	1	CRV	31495

Reaction No. 80077							
$2\text{PbSbO}_2\text{Cl}(\text{s}) + 2\text{H}^+ = \text{Sb}_2\text{O}_3(\text{s}) + 2\text{Pb}^{2+} + 2\text{Cl}^- + \text{H}_2\text{O}$							
Note: Mineral name: nadorite							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1084	HNO3	lgK:-12.36(4SF)	4	MSL	29879
Note (1): Authors also give deltaG of formation							

Reaction No. 80078							
$\text{NaSbO}_3(\text{s}) + 3\text{H}_2\text{O} = \text{Sb}^{5+}\text{OH}^-(6) + \text{Na}^+$							
Note: Mineral name: brizziite							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1491	CO3-2 salt	lgK:-6.70(0.10SD)	4	MSL	31678

Reaction No. 80079							
$2.5\text{H}_2(\text{g}) + 2.5\text{O}_2(\text{g}) + \text{Sb}(\text{s}) = \text{Sb}^{5+}\text{OH}^-(5)$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:170.5(4SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-996.6(1.0SD)kJ	0	EOD	30052

Reaction No. 80158							
$\text{Sb}^{5+}\text{OH}^-(6) + \text{H}^+(2)\text{Citric-3} = \text{Sb}^{5+}\text{H}^+\text{OH}^-(4)\text{Citric-3} + \text{OH}^- + \text{H}_2\text{O}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	KNO3	lgK:-6.5(0.4SD)	2	MGL	25244
Note (1): Unbalanced rxn corrected							

Reaction No. 80192							
$\text{Sb}^{5+}\text{OH}^-(6) + \text{H}^+\text{Oxalic-2} = \text{Sb}^{5+}\text{OH}^-(4)\text{Oxalic-2} + \text{H}_2\text{O} + \text{OH}^-$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	KNO3	lgK:-6.4(2SF)	3	MGL	23086
Note (1): Reaction corrected.							
2	25	0.1	KNO3	lgK:-6.2(0.4SD)	5	MGL	25244

Reaction No. 80193							
$\text{Sb}^{5+}\text{OH}^-(6) + \text{H}^+\text{Salicylic-2} = \text{Sb}^{5+}\text{OH}^-(4)\text{Salicylic-2} + \text{H}_2\text{O} + \text{OH}^-$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	KNO3	lgK:-2.7(0.6SD)	5	MGL	25244

Reaction No. 80194							
$\text{Sb+5_OH-1(6)} + \text{H+1_Malonic-2} = \text{Sb+5_OH-1(4)_Malonic-2} + \text{H2O} + \text{OH-1}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	KNO3	lgK:-6.9(2SF)	3	MGL	25244
Note (1): <-6.9							

Reaction No. 80195							
$\text{Sb+5_OH-1(6)} + \text{H+1_Gly-1} = \text{Sb+5_Gly-1_OH-1(4)} + \text{H2O} + \text{OH-1}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	KNO3	lgK:-7.3(2SF)	3	MGL	25244
Note (1): <-7.3							

Reaction No. 80219							
$\text{Sb+5_OH-1(6)} + \text{Citric-3} + \text{H+1} = \text{Sb+5_H+1(-1)_OH-1(4)_Citric-3} + 2<\text{H2O}>$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	KNO3	lgK:8.0(0.4SD)	4	MGL	25244

Reaction No. 80220							
$2<\text{SbO+1}> + 3<\text{H+1(2)_S-2(g)}> = \text{Sb+3(2)_S-2(3)_(s)} + 2<\text{H+1}> + 2<\text{H2O}>$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-38.(2SF)	0	CRV	600
Note (1): SbO+ does not exist as such; it would be possible to rewrite as Sb(OH)2							

Reaction No. 80811							
$\text{FeSb2O4(s)} + 2<\text{H+1}> = \text{Fe+2} + \text{Sb2O3(s)} + \text{H2O}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:0.155(3SF)	4	MSL	30052
Note (1): Authors also give deltaG of formation; Sb2O3(s) is senarmontite							

Reaction No. 80812							
$\text{MgSb2O6(s)} + 4<\text{H2O}> + 2<\text{H+1}> = \text{Mg+2} + 2<\text{Sb+5_OH-1(5)}>$							
Note: Mineral name: bystroemite							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1053	HNO3	lgK:-9.44(0.10SD)	4	MSL	30052
Note (1): Authors also give deltaG of formation							

Reaction No. 80813							
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PbSb2O6(s) + 4<H2O> + 2<H+1> = Pb+2 + 2<Sb+5_OH-1(5)>							
Note: Mineral name: rosiaite							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.0996	HNO3	lgK:-14.92(0.10SD)	4	MSL	30052
Note (1): Authors also give deltaG of formation							

Reaction No. 80814							
ZnSb2O6(s) + 4<H2O> + 2<H+1> = Zn+2 + 2<Sb+5_OH-1(5)>							
Note: Mineral name: ordonezite							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1053	HNO3	lgK:-11.37(0.10SD)	4	MSL	30052
Note (1): Authors also give deltaG of formation							

Reaction No. 80815							
Sb+5_OH-1(6) + 3<H+1(2)_Cat-2> = Sb+5_Cat-2(3) + 6<H2O>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	KNO3	lgK:5.5(0.5SD)	4	MGL	25244

Reaction No. 80816							
Sb+5_OH-1(6) + Lactic-1 = Sb+5_H+1(-1)_Lactic-1_OH-1(4) + H2O + OH-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	KNO3	lgK:-6.6(0.3SD)	4	MGL	25244

Reaction No. 80817							
Sb+5_OH-1(6) + 3<Xylitol> = Sb+5_H+1(-6)_Xylitol(3) + 6<H2O>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	KNO3	lgK:5.6(0.5SD)	4	MGL	25244

Reaction No. 80818							
Sb+5_OH-1(6) + 3<DMannitol> = Sb+5_H+1(-6)_DMannitol(3) + 6<H2O>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	KNO3	lgK:5.6(0.5SD)	4	MGL	25244

Reaction No. 80823							
0.5<F2(g)> + 0.5<O2(g)> + Sb(s) = SbO+1_F-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-487.4(4SF)kJ	1	EFE	9015

2	25	0	Inf. Dilution	dG:-116.5(4SF)	4	CRV	12011
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Reaction No. 80824							
2.5<Cl ₂ (g)> + Sb(s) = SbCl ₅ (g)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-334.29(5SF) kJ	3	EFE	9015
2	25	0	Inf. Dilution	dG:-79.91(4SF)	3	EFE	31675

Reaction No. 80825							
Sb+3(2)_S-2(3)_(s) + 2<H+1_S-2> = Sb+3(2)_H+1(2)_S-2(5)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-2.56(3SF)	1	EES	
2	30	0	Self Medium	lgK:-2.56(0.04SD)	3	MSL	29949

Reaction No. 80826							
0.5<Sb+3(2)_S-2(3)_(s)> + 3<H ₂ O> = Sb+3_OH-1(3) + 1.5<H+1_S-2> + 1.5<H+1 >							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	30	0	Self Medium	lgK:-26.5(0.1SD)	3	MSL	29949

Reaction No. 80827							
Sb+3_OH-1(3) + 3<H+1> + Cl-1 = Sb+3_Cl-1 + 3<H ₂ O>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:2.78(3SF)	1	EES	
2	25	0	Inf. Dilution	lgK:2.78(3SF)	2	CRV	31301
3	25	0	Inf. Dilution	lgK:2.78(3SF)	2	CRV	31495

Reaction No. 80828							
Sb+3_OH-1(3) + 3<H+1> + 2<Cl-1> = Sb+3_Cl-1(2) + 3<H ₂ O>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:4.86(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:3.27(3SF)	0	CRV	31301
3	25	0	Inf. Dilution	lgK:3.27(3SF)	0	CRV	31495

Reaction No. 80829							
Sb+3 + H+1_F-1 = Sb+3_F-1 + H+1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:0.15(2SF)	1	ELC	

2	25	0	Inf. Dilution	lgK:4.03(3SF)	0	CRV	31301
3	25	0	Inf. Dilution	lgK:4.03(3SF)	0	CRV	31495
Note (3): Authors used [437] data but writing HF instead of F							

Reaction No. 80830							
$\text{Sb}+3 + 2\langle\text{H}+1_F-1\rangle = \text{Sb}+3_F-1(2) + 2\langle\text{H}+1\rangle$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-0.59(2SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:7.02(3SF)	0	CRV	31301
3	25	0	Inf. Dilution	lgK:7.02(3SF)	0	CRV	31495
Note (3): Authors used [437] data but writing HF instead of F							

Reaction No. 80831							
$\text{Sb}+3 + 3\langle\text{H}+1_F-1\rangle = \text{Sb}+3_F-1(3) + 3\langle\text{H}+1\rangle$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-1.19(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:9.55(3SF)	0	CRV	31301
3	25	0	Inf. Dilution	lgK:9.55(3SF)	0	CRV	31495
Note (3): Authors used [437] data but writing HF instead of F							

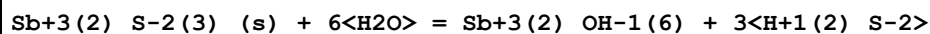
Reaction No. 80832							
$\text{Sb}+3_F-1(3) + \text{H}_2\text{O} = \text{Sb}+3_F-1(3)_OH-1 + \text{H}+1$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.003:0.1	Self Medium	lgK:-3.92(3SF)	3	MPT	16752

Reaction No. 80833							
$2\langle\text{Sb}+3_F-1(3)\rangle + \text{H}_2\text{O} = \text{Sb}+3_F-1(2) + \text{F}-1 + \text{Sb}+3_F-1(3)_OH-1 + \text{H}+1$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.003:0.1	Self Medium	lgK:-2.82(3SF)	0	MRM	16752
2	25	0	Inf. Dilution	lgK:-6.9(2SF)	1	ELC	

Reaction No. 80834							
$0.5\langle\text{Sb}+3(2)_S-2(3)_s\rangle + 3\langle\text{H}_2\text{O}\rangle = \text{Sb}+3_OH-1(3) + 1.5\langle\text{H}+1(2)_S-2\rangle$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-20.3(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:-17.011(5SF)	0	EOD	31708
3	50	0	Inf. Dilution	lgK:-15.329(5SF)	0	EOD	31708

4	100	0	Inf. Dilution	lgK:-12.561(5SF)	0	EOD	31708
5	150	0	Inf. Dilution	lgK:-10.385(5SF)	0	EOD	31708
6	200	0	Inf. Dilution	lgK:-8.626(4SF)	0	EOD	31708
7	250	0	Inf. Dilution	lgK:-7.150(4SF)	0	EOD	31708
8	300	0	Inf. Dilution	lgK:-5.902(4SF)	0	EOD	31708
9	350	0	Inf. Dilution	lgK:-4.684(4SF)	0	EOD	31708

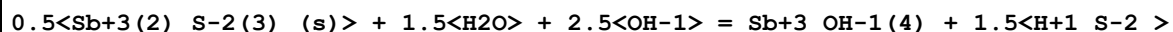
Reaction No. 80835



Note: Sb₂(OH)₆ only proposed in 31708 on the basis of "estimations"; dimer maybe relevant above 250C

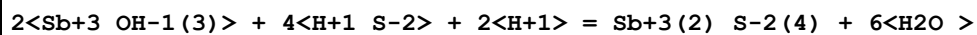
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-33.946(5SF)	0	EOD	31708
2	50	0	Inf. Dilution	lgK:-30.415(5SF)	0	EOD	31708
3	100	0	Inf. Dilution	lgK:-24.617(5SF)	0	EOD	31708
4	150	0	Inf. Dilution	lgK:-20.065(5SF)	0	EOD	31708
5	200	0	Inf. Dilution	lgK:-16.388(5SF)	0	EOD	31708
6	250	0	Inf. Dilution	lgK:-13.307(5SF)	0	EOD	31708
7	300	0	Inf. Dilution	lgK:-10.705(5SF)	0	EOD	31708
8	350	0	Inf. Dilution	lgK:-8.179(4SF)	0	EOD	31708

Reaction No. 80836



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-7.6(2SF)	1	ELC	31708
2	25	0	Inf. Dilution	lgK:-4.436(4SF)	0	EOD	31708
3	50	0	Inf. Dilution	lgK:-3.544(4SF)	0	EOD	31708
4	100	0	Inf. Dilution	lgK:-2.130(4SF)	0	EOD	31708
5	150	0	Inf. Dilution	lgK:-1.072(4SF)	0	EOD	31708
6	200	0	Inf. Dilution	lgK:-0.2407(4SF)	0	EOD	31708
7	250	0	Inf. Dilution	lgK:0.361(3SF)	0	EOD	31708
8	300	0	Inf. Dilution	lgK:0.979(3SF)	0	EOD	31708
9	350	0	Inf. Dilution	lgK:1.431(4SF)	0	EOD	31708

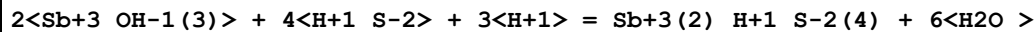
Reaction No. 80837



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
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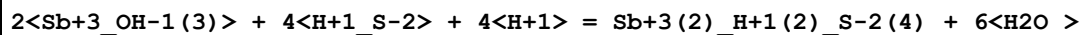
1	25	0	Inf. Dilution	lgK:49.4 (3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:42.53 (4SF)	0	CRV	31301
3	25	0	Inf. Dilution	lgK:42.53 (4SF)	0	CRV	31495

Reaction No. 80838



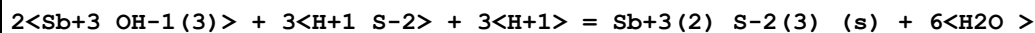
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:60.0 (3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:52.18 (4SF)	0	CRV	31301
3	25	0	Inf. Dilution	lgK:52.18 (4SF)	0	CRV	31495

Reaction No. 80839



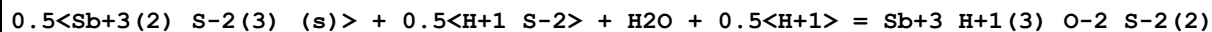
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:63.3 (3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:57.00 (4SF)	0	CRV	31301
3	25	0	Inf. Dilution	lgK:57.00 (4SF)	0	CRV	31495

Reaction No. 80840



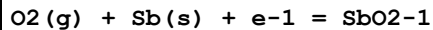
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:61.5 (3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:55.14 (4SF)	0	CRV	31301
3	25	0	Inf. Dilution	lgK:55.14 (4SF)	0	CRV	31495

Reaction No. 80841



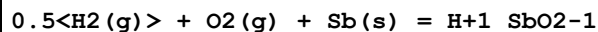
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	30	0	Self Medium	lgK:-4.32 (3SF)	2	MSL	29949

Reaction No. 81144



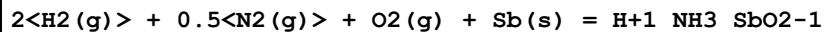
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-81.32 (4SF)	4	CRV	12011

Reaction No. 81145



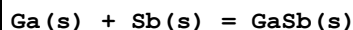
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-97.4 (3SF)	4	CRV	12011
2	25	0	Inf. Dilution	dH:-116.6 (4SF)	4	CRV	12011

Reaction No. 81146



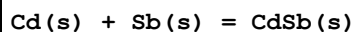
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-100.29 (5SF)	4	CRV	12011

Reaction No. 81192



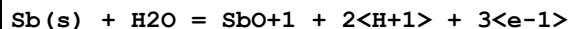
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-9.3 (2SF)	4	CRV	12011
2	25	0	Inf. Dilution	dH:-10.0 (3SF)	4	CRV	12011

Reaction No. 81280



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-3.11 (3SF)	4	CRV	12011
2	25	0	Inf. Dilution	dH:-3.44 (3SF)	4	CRV	12011

Reaction No. 81576



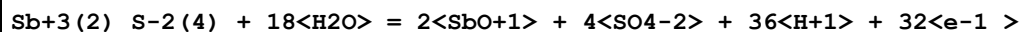
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-10.51 (4SF)	4	CRV	32224

Reaction No. 81577



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-159.8 (4SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:-158.64 (5SF)	0	CRV	32224

Reaction No. 81578



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-181.42 (5SF)	4	CRV	32224

Reaction No. 81579

Sb+3_Cl-1(3)_(s) + H2O = SbO+1 + 2<H+1> + 3<Cl-1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.763(4SF)	4	CRV	32224

Reaction No. 81580							
Sb+3_Br-1(3)_(s) + H2O = SbO+1 + 2<H+1> + 3<Br-1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:2.21(3SF)	4	CRV	32224

Reaction No. 81581							
Sb+3_F-1(3)_(s) + H2O = SbO+1 + 2<H+1> + 3<F-1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-12.0(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:-10.44(4SF)	0	CRV	32224

Reaction No. 81582							
SbO+1_F-1 = SbO+1 + F-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-5.51(3SF)	4	CRV	32224

Reaction No. 81583							
Sb+3_F-1_OH-1(2) = SbO+1 + F-1 + H2O							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-5.15(3SF)	4	CRV	32224

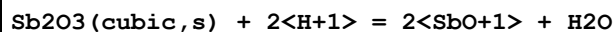
Reaction No. 81584							
SbO2-1 + 2<H+1> = SbO+1 + H2O							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:12.97(4SF)	4	CRV	32224

Reaction No. 81585							
H+1_SbO2-1 + H+1 = SbO+1 + H2O							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.19(3SF)	4	CRV	32224

Reaction No. 81586							
Sb2O3(rhombic,s) + 2<H+1> = 2<SbO+1> + H2O							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

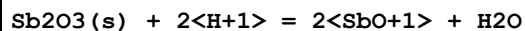
1	25	0	Inf. Dilution	lgK:-6.20 (3SF)	4	CRV	32224
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Reaction No. 81587



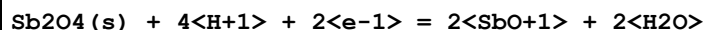
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-6.5 (2SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:-3.05 (3SF)	0	CRV	32224

Reaction No. 81588



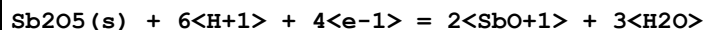
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-5.3 (2SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:-2.32 (3SF)	0	CRV	32224

Reaction No. 81589



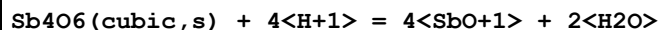
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:5.73 (3SF)	4	CRV	32224

Reaction No. 81590



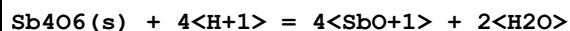
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:40.5 (3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:41.42 (4SF)	0	CRV	32224

Reaction No. 81591



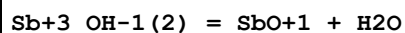
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-15.9 (3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:-14.97 (4SF)	0	CRV	32224

Reaction No. 81592



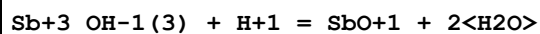
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-12.33 (4SF)	4	CRV	32224

Reaction No. 81593



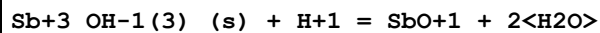
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-0.20 (2SF)	4	CRV	32224

Reaction No. 81594



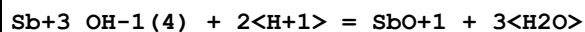
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.18 (3SF)	4	CRV	32224

Reaction No. 81595



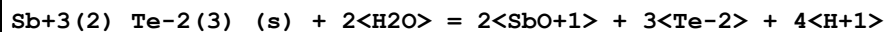
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.9 (2SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:-5.95 (3SF)	0	CRV	32224

Reaction No. 81596



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:12.97 (4SF)	4	CRV	32224

Reaction No. 81597



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-147.2 (4SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:-142.13 (5SF)	0	CRV	32224

Data Assessment

Weight	Assessment
9	highest accuracy
8	multi-determined; excellent dependability
7	trustworthy; outstanding single determination
6	better than average reliability
5	average single determination
4	some doubt
3	poorly known
2	guess to achieve consistency (or just as bad)

1	no information but consistent
0	problematic/inconsistent

Techniques

Abbrev	Method
CRV	Critical review
CSM	Smith & Martell Vols 1-6 (crit.)
EAE	Automated extrapolation
EBP	By Proxy
EDH	Debye-Hueckel corrections
EES	Personal estimate
EFE	Free Energy relations
EIU	Critical IUPAC review (estim.)
ELC	Linear Combination of Reactions
EMU	Method unknown
ENS	Not specified
EOD	Calculated from data of others
EOR	Other Reaction Constants
ESC	Standard conditions anchor
ETR	Thermodynamic relationships
MCL	Calorimetry
MDG	Combination of Thermodynamic Data
MDS	Distribution between two phases
MEF	Electromotoric force, not specified
MGL	Glass electrode
MPH	pH method, not specified
MPL	Polarography
MPS	Potentiometry and Spectrophotometry
MPT	Potentiometry
MRM	Raman spectra
MSL	Solubility
MSP	Spectroscopy
MTD	Temperature difference

MVM	Voltametric Methods
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